CEBU INSTITUTE OF TECHNOLOGY UNIVERSITY

COLLEGE OF COMPUTER STUDIES

Software Requirements Specifications

for

IntelliTrack

Change History

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1. Introduction

1.1. Purpose

Capstone projects often fail due to students' poor time management, disorganized submissions, and last-minute work across multiple deliverables. Faculty advisers also struggle with manually tracking various groups, monitoring deadlines, reviewing submissions, and providing timely feedback. Existing platforms such as LMSs and academic portals offer only basic file submission and storage. They lack milestone-based tracking, automated reminders, version control, risk detection, and structured feedback loops specifically designed for capstone workflows.

The purpose of this document is to serve as a reference for developers, project advisers, and stakeholders by clearly describing how the system should behave, what features it must include, and the constraints under which it operates.

IntelliTrack aims to improve capstone project workflows by providing:

- Automated milestone tracking
- AI-assisted deadline reminders
- Centralized document submission
- Real-time monitoring of deliverables
- Analytics dashboards for instructors and coordinators

By implementing these features, the system enhances student accountability, reduces administrative workload, and improves overall capstone project management.

1.2. Scope

IntelliTrack is a web-based capstone deliverables submission tracking system designed to help academic institutions monitor student progress through time-based submission tracking and data analytics.

The system focuses on improving deadline compliance, submission monitoring, and data-driven decision making by providing tools that allow students, advisers, coordinators, and administrators to track capstone deliverables in real time.

Through IntelliTrack, students can upload project deliverables and monitor submission deadlines, while advisers and coordinators can analyze submission behavior using system-generated analytics dashboards.

The system emphasizes time monitoring and submission analytics by tracking submission timestamps, identifying late or missing deliverables, and generating insights that help advisers detect at-risk students or groups.

Key capabilities include:

- Secure Gmail-based user authentication for role-based access
- Capstone class and deliverable configuration by administrators
- Automated tracking of submission deadlines and timestamps
- Real-time monitoring of submission status (submitted, pending, late)
- AI-assisted reminders to encourage timely submissions
- Data analytics dashboards that visualize submission trends and compliance rates

The primary focus of IntelliTrack is time-based tracking of capstone deliverables and analytical monitoring of submission performance. The system is not intended to replace full Learning Management Systems (LMS), but rather to serve as a specialized monitoring and analytics tool for capstone project management.

1.3. Definitions, Acronyms and Abbreviations

Acronym & Abbreviations	**Definition**
AI	Artificial Intelligence
RBAC	Role-Based Access Control
LMS	Learning Management System
UI	User Interface

UX	User Experience
CRUD	Create, Read, Update, Delete (basic operations in software systems)
CDSS	Capstone Deliverables Submission System
Milestone	A required capstone deliverable with a specific deadline (e.g., Proposal, Midterm, Final)
Version Control	A system that tracks document versions, changes, and submission histories
Analytics Dashboard	A visual interface showing real-time submission status and progress insights
Inline Feedback	Comments or annotations placed directly on a student's submitted document

1.4. References

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Source: https://learningmole.com/ai-powered-feedback-systems-for-enhanced-learning/

- **Dönmez, M.** (2024). *AI-Based Feedback Tools in Education: A Comprehensive Bibliometric Analysis Study.* International Journal of Assessment Tools in Education, 11(4), 622–646.
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  Source: https://www.gse.harvard.edu/ideas/usable-knowledge/23/11/better-feedback-ai

2. Overall Description

2.1. Product perspective

Capstone projects require students to complete multiple deliverables within strict timelines. However, many institutions rely on manual tracking methods or general learning management systems that lack dedicated tools for monitoring submission timing and analyzing student compliance with deadlines.

IntelliTrack is designed as a **specialized capstone submission tracking and analytics system** that focuses on monitoring deliverables based on **time and submission behavior**. The system provides automated deadline monitoring, submission timestamp tracking, and data analytics dashboards that allow advisers and coordinators to evaluate the progress of students and groups throughout the capstone process.

The system operates as a **cloud-based web application using a client–server architecture**. Users access the system through a web browser while backend services handle authentication, submission tracking, time monitoring, and analytics processing.

IntelliTrack integrates with external services such as **Google Authentication for secure login and Google Sheets for structured data synchronization and backup**. By combining submission tracking with data analytics, the system provides real-time insights into submission performance and deadline compliance.

2.2. User characteristics

The IntelliTrack system provides several core functions designed to support **capstone deliverable tracking and analytics monitoring**.

- **User Authentication and Role Management**

The system authenticates users through institutional Gmail accounts and assigns roles such as student, adviser, coordinator, or administrator.

- **Class and Deliverable Management**

  Administrators can create classes and define deliverables with specific deadlines to structure the capstone workflow.

- **Submission Tracking**

  Students can upload deliverables while the system records submission timestamps and automatically determines whether submissions are on time, pending, or late.

- **Deadline Monitoring**

  The system continuously monitors deadlines and generates reminders to help students submit deliverables on time.

- **Time-Based Analytics**

  Submission data is analyzed to generate insights such as submission trends, late submission rates, and overall compliance with deadlines.

- **Analytics Dashboard**

  Advisers and coordinators can access dashboards that visualize submission statistics and provide insights into student and group performance.

2.4. Constraints

The development and deployment of **IntelliTrack** are subject to the following constraints:

Regulatory Policies

- Must comply with institutional academic policies
- Must follow data privacy regulations such as Data Privacy Act of 2012 (Philippines)
- FERPA/GDPR alignment when applicable

Hardware & Software Limitations

- Requires stable internet access
- Dependent on availability of cloud hosting
- Performance may vary with large file uploads

Interface Constraints

- Must integrate with institutional email systems
- Must support document formats (PDF, DOCX, PPTX)
- Connection with cloud storage API

Parallel Operation

- Must support simultaneous access from multiple users (students, advisers, admins)
- Should maintain performance at 500-1000 and more concurrent users

Audit & Logging Requirements

- Must maintain audit trails for all submissions, edits, role changes, and feedback
- Logs should be immutable for academic integrity

Control Functions

- Version control must prevent document overwriting
- System must enforce role-based restrictions

Reliability Requirements

- Target uptime: **99.9% availability**
- Automated backups and recovery plans required

Criticality

- High academic impact: failure may affect grading and submission deadlines
- Must ensure reliable access, especially near milestone deadlines

Safety & Security Considerations

- Encrypted document storage
- Role-based access control
- Anomaly detection for suspicious activity
- Secure authentication for faculty accounts

2.5. Assumptions and dependencies

Assumptions

The following conditions are assumed to be true for the **IntelliTrack** to operate as intended:

- Users have stable internet access and use modern browsers (Chrome, Edge, Firefox, Safari).
- Institutional administrators will provide accurate academic schedules, deadlines, user lists, and role assignments.
- Students and faculty are familiar with uploading and downloading digital documents.
- Institutional email services are available and functioning for authentication and account verification.
- Cloud hosting services (e.g., Azure, AWS) remain operational and available.
- All users access the system with their assigned institution-based accounts.
- Advisers and coordinators provide timely feedback and updates to ensure analytics remain accurate.
- Document uploads follow accepted formats (PDF, DOCX, PPTX) as defined by institutional policies.
- Users have devices capable of accessing the system (PC, laptop, or mobile device).
- File sizes uploaded by students are within allowed storage limitations.
- System maintenance schedules are coordinated and communicated in advance to

avoid academic disruptions.

- Institutional policies regarding data retention, privacy, and academic conduct remain stable throughout development.

Dependencies

The **IntelliTrack** relies on the following external systems, technologies, and conditions:

- Availability and reliability of the cloud database (e.g., Supabase) for storing submissions, logs, and metadata.
- AI-generated reminders and risk detection depend on accurate and complete data inputs (deadlines, submission timestamps, user activity).
- Document management, version control, and rollback features rely on third-party storage APIs and cloud file systems.
- Real-time analytics and reporting depend on uninterrupted logging services and complete submission records.
- Security features depend on the hosting provider's infrastructure for encryption, secure authentication, and intrusion detection.
- Email notification services must remain available.
- Integration with institutional systems (e.g., student information systems, email servers) must remain functional.
- System performance depends on the scalability and responsiveness of the hosting platform.
- Mobile access depends on device compatibility and browser support for responsive UI components.
- Backup and recovery procedures depend on cloud provider backup services.
- Future upgrades may depend on API updates or changes from third-party providers.
- The development team relies on version control tools (e.g., GitHub) for stable collaboration and deployment pipelines.

3. Specific Requirements

3.1. External interface requirements

3.1.1. Hardware interfaces

Client Devices:

- **Devices Supported:** The system must be accessible on any device with a modern browser, including PC, laptop, or mobile device (though the UI/UX is tailored for desktop compatibility).
- **User Interface (UI) Support:** The system requires full-screen support and must render a simple, intuitive interface with easy navigation across different screen resolutions (responsive design is implied for cross-platform compatibility).
- **Input/Output:** Standard input devices (keyboard, mouse/touchpad) are required for login, document upload, and inputting inline comments/rubric evaluations.

Deployment & Hosting:

- **Server Hardware:** The system requires Cloud hosting with database support.
- **Performance Configuration:** The underlying server hardware must be provisioned to handle at least 1,000 concurrent users and maintain a response time of < 2 seconds.

3.1.2. Software interfaces

Required Software Products (Core Stack):

- **Operating System:** Any standard operating system that supports the required modern web browsers.
- **Data Management System:** The system is dependent on the availability and reliability of a cloud database for structured storage of submissions, logs, and metadata.
- **Backend Framework: Django** is required for secure, modular business logic.
- **Frontend Framework: React.jsx** is required for the user interface.
- **Version Control:** The development team relies on **version control tools (e.g.,**

GitHub) for stable collaboration.

Possible Interfaces with Other Application Systems:

- **Institutional Email System:** The system must **integrate with institutional email systems** for **account verification**, login, and sending **automated reminders (via email)**.
- **Cloud Storage Services:** The system must connect with **cloud storage API** for **secure file handling**, managing **document version control**, and **rollback features**.
- **AI/Analytics Modules:** Interfaces with **AI modules** for **risk prediction**, **smart reminders**, and **analytics engines** for **progress tracking and reporting**.

3.1.3. Communications interfaces

Network Access: The system requires **stable internet access** as it is a web-based, cloud-hosted application.

Protocols and Security:

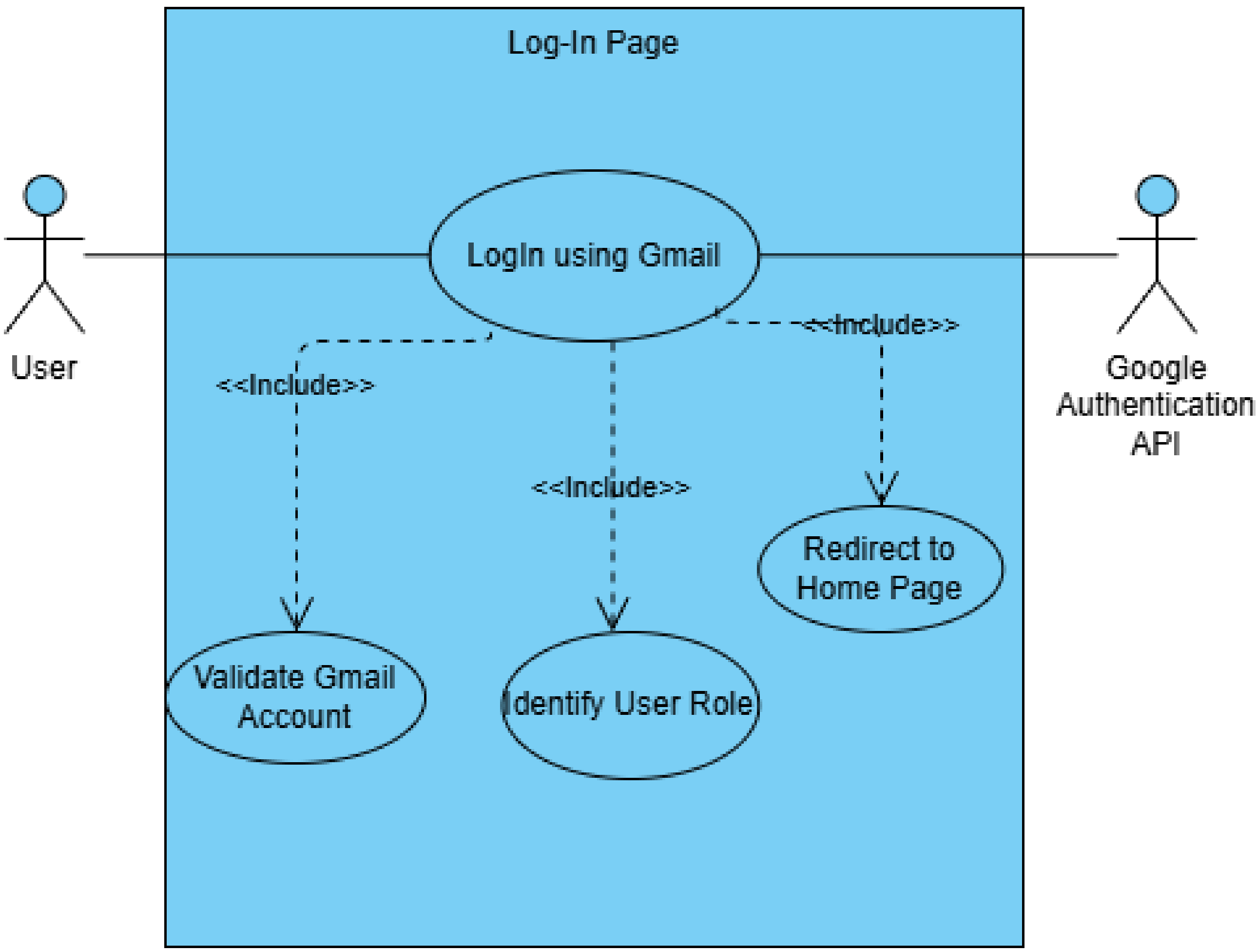
- **Data Encryption:** Requires protocols to ensure **end-to-end encryption for stored and transmitted data**.
- **Authentication Protocol:** Must integrate with institutional authentication protocols to use **institutional email** and support **Multi-Factor Authentication (MFA)** for advisers and administrators.
- **Email Protocol:** The system relies on **Email notification services (SMTP or third-party services)** for sending alerts and reminders.
- **API Communication:** Communication protocols are needed for the **connection with cloud storage API** and other external services (analytics, institutional systems).

3.2. Functional requirements

Module 1: User Management System

1.1 Login

- *Use Case Diagram*

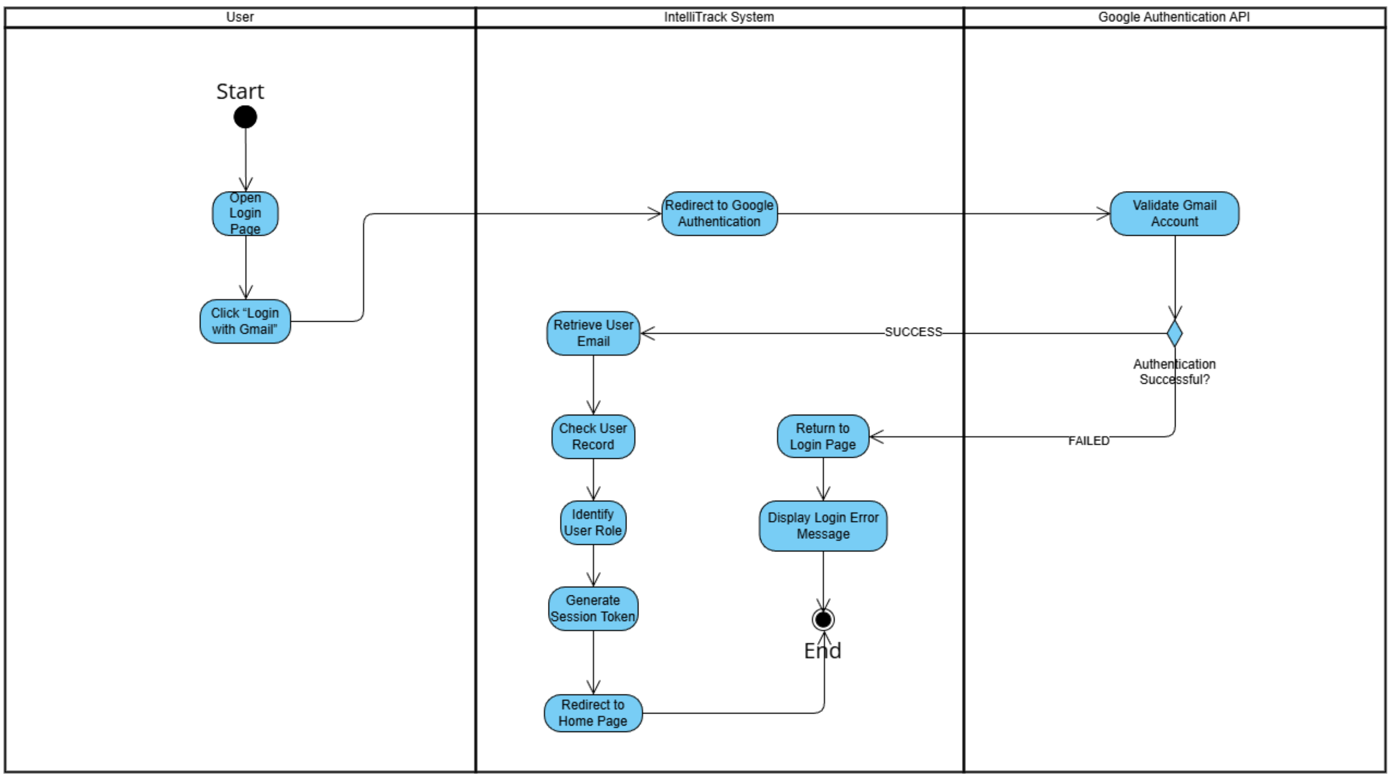


- *Use Case Description*

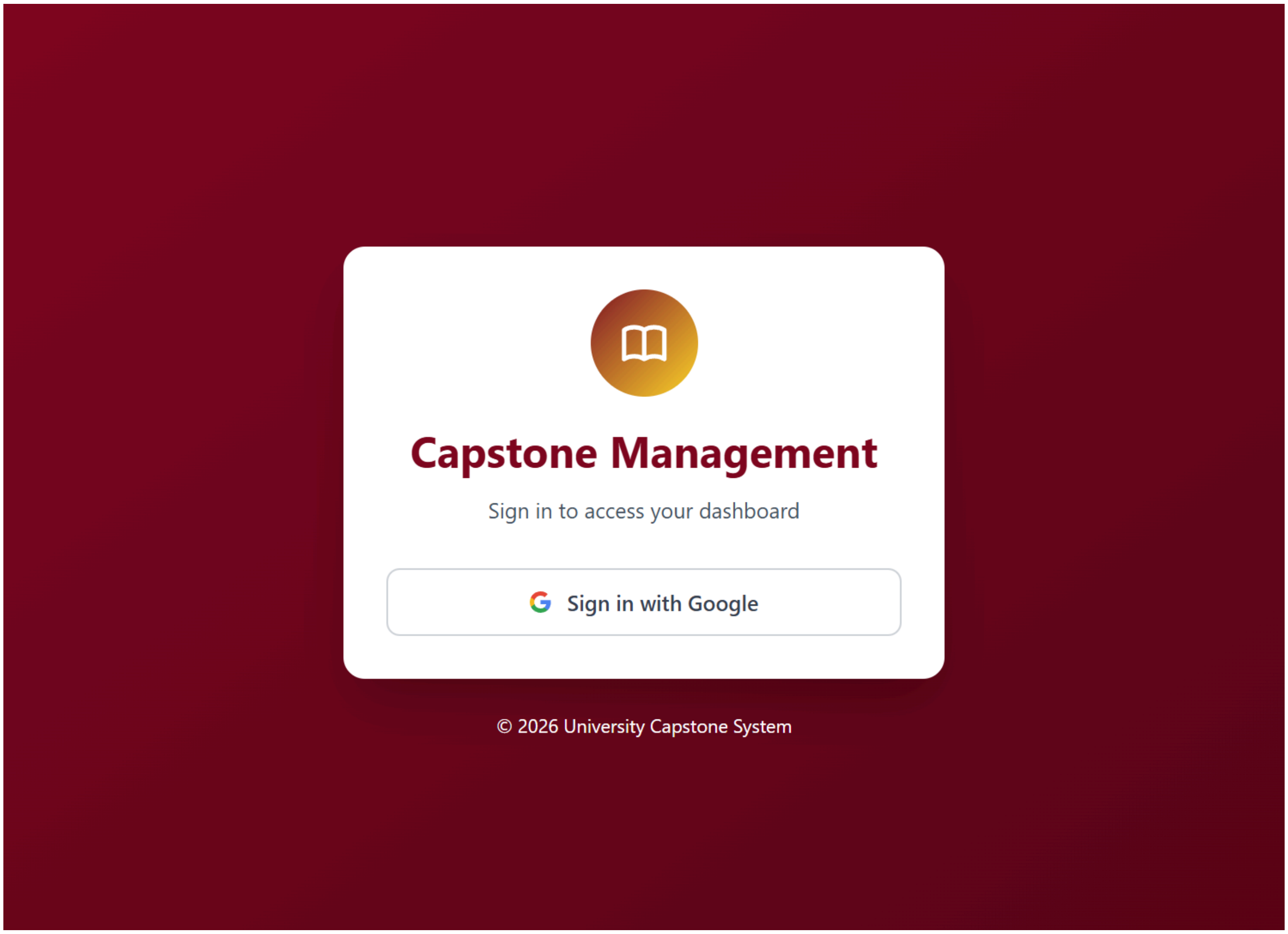
The system shall allow users to authenticate using their institutional Gmail accounts. When the user initiates the login process, the system shall redirect the user to Google OAuth authentication to verify their identity. Upon successful authentication, the system shall validate the user’s email and retrieve the corresponding user record from the database.

The system shall then identify the user's assigned role (Student, Adviser, Coordinator, or Administrator) and generate a secure session. Based on the identified role, the user shall be redirected to a personalized dashboard. If authentication fails, the system shall display an error message and allow the user to retry.

- _Activity Diagram_

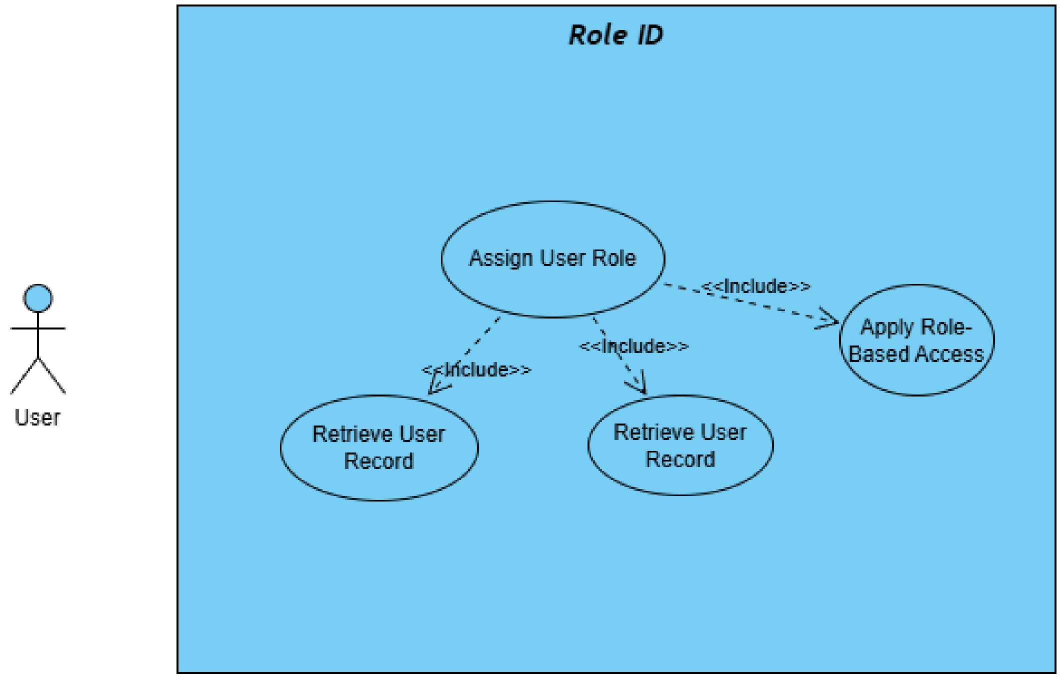


- *Wireframe*


Capstone Management
Sign in to access your dashboard
Sign in with Google
© 2026 University Capstone System

1.2 Role Identification

- *Use Case Diagram*

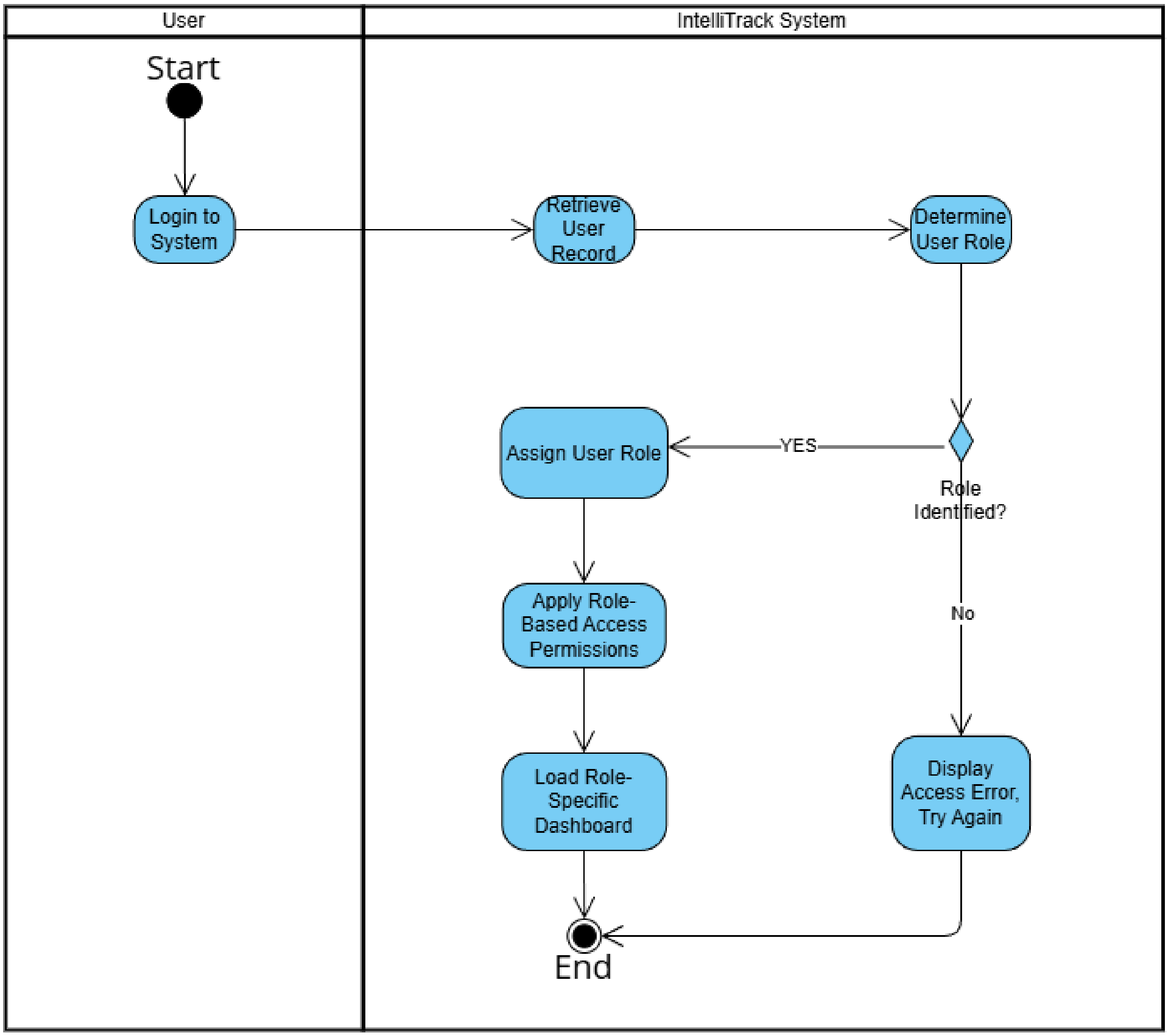


- *Use Case Description*

The system shall automatically determine and assign user roles based on stored user and class records during the authentication process. After a successful login, the system shall retrieve the user's information and identify the appropriate role, such as Student, Adviser, Coordinator, or Administrator.

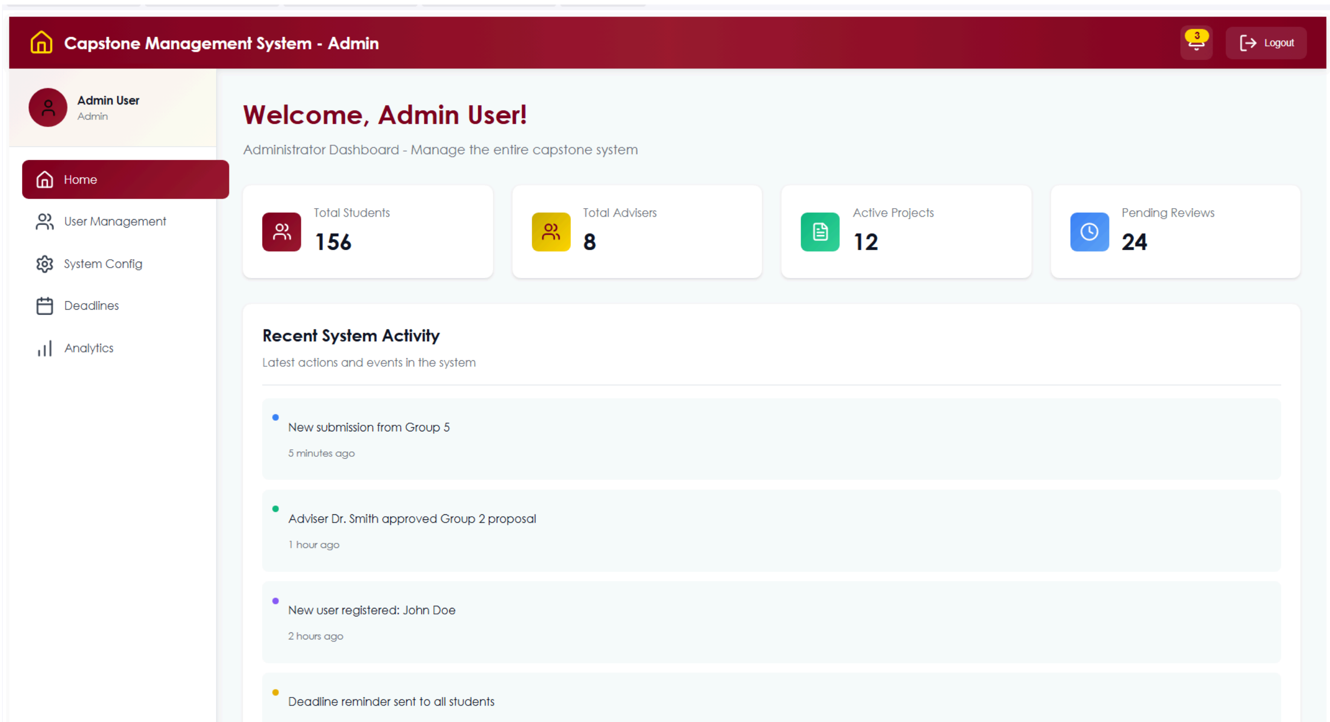
The system shall enforce role-based access control by restricting or granting access to system features according to the assigned role. Each user shall only be able to access modules and data relevant to their responsibilities within the system.

- *Activity Diagram*

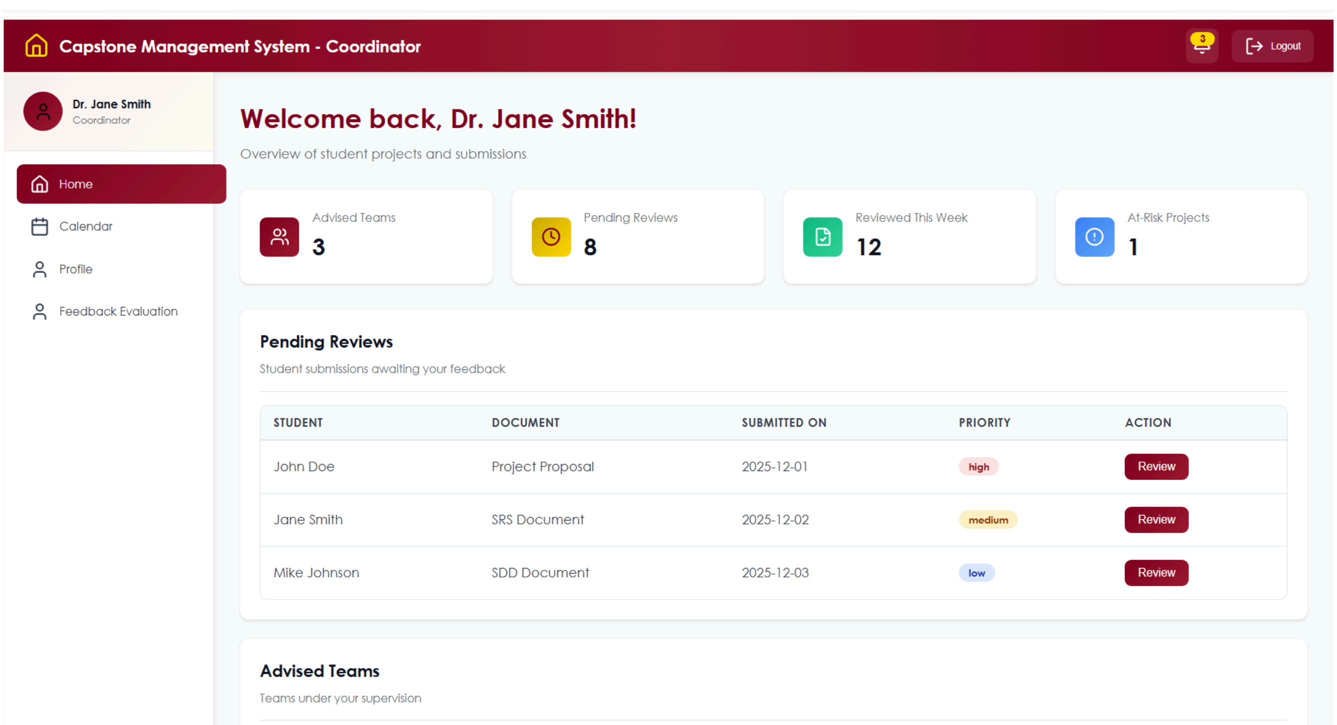


- *Wireframe*

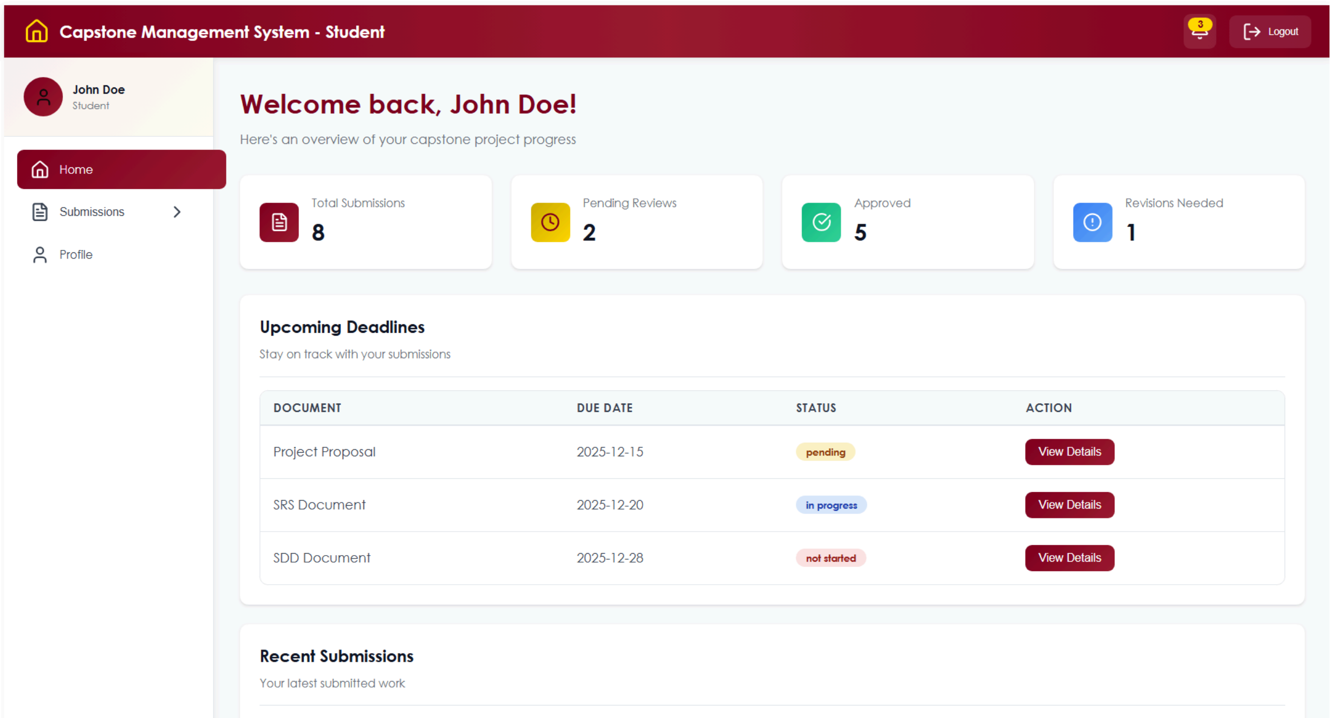
Admin



Coordinator

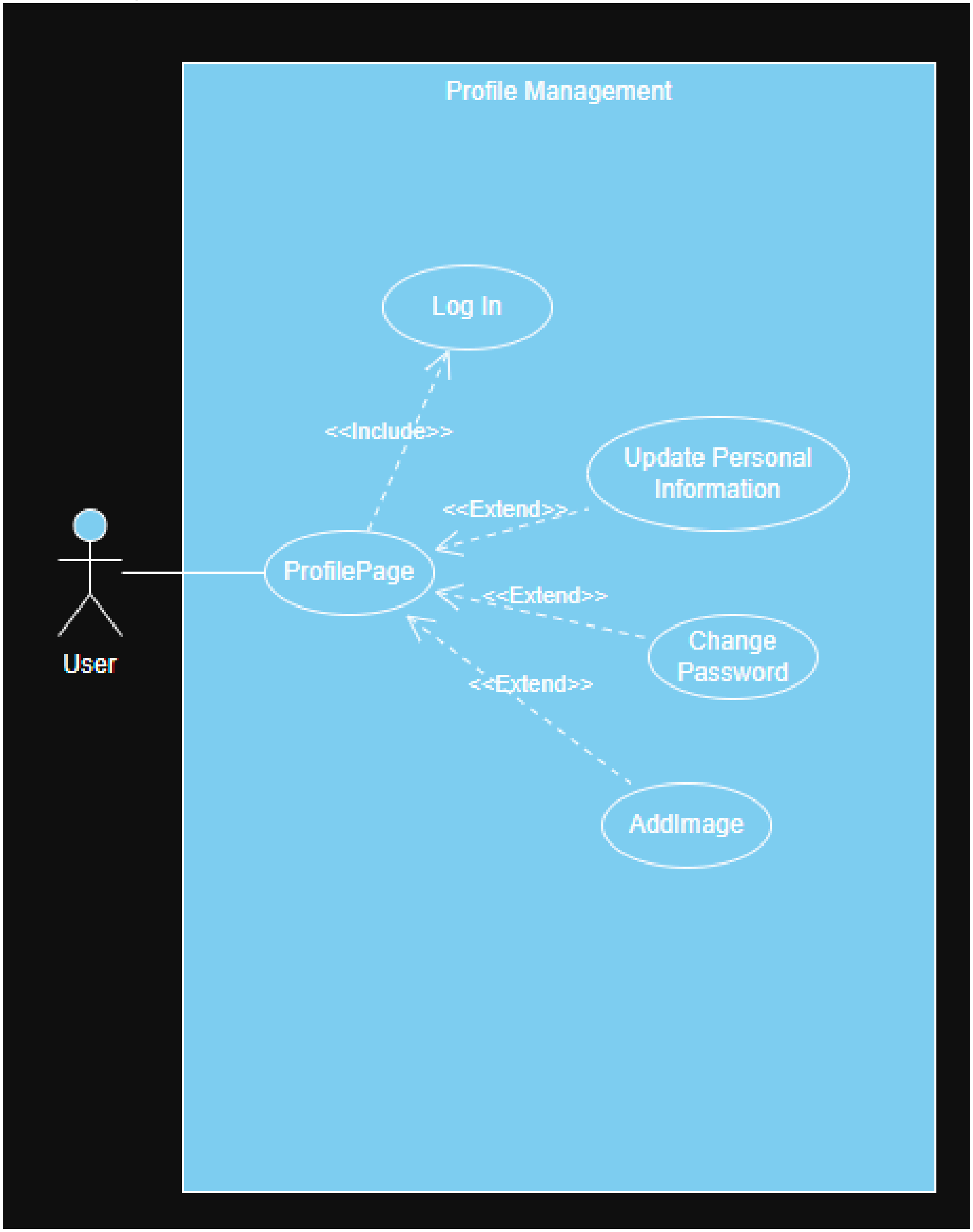


Student


Capstone Management System - Student
3
Logout
John Doe
Student
Home
Submissions
Profile
Welcome back, John Doe!
Here's an overview of your capstone project progress
Total Submissions
8
Pending Reviews
2
Approved
5
Revisions Needed
1
Upcoming Deadlines
Stay on track with your submissions
DOCUMENT
DUE DATE
STATUS
ACTION
Project Proposal
2025-12-15
pending
View Details
SRS Document
2025-12-20
in progress
View Details
SDD Document
2025-12-28
not started
View Details
Recent Submissions
Your latest submitted work

1.3 **Profile Management**

- _Use Case Diagram_

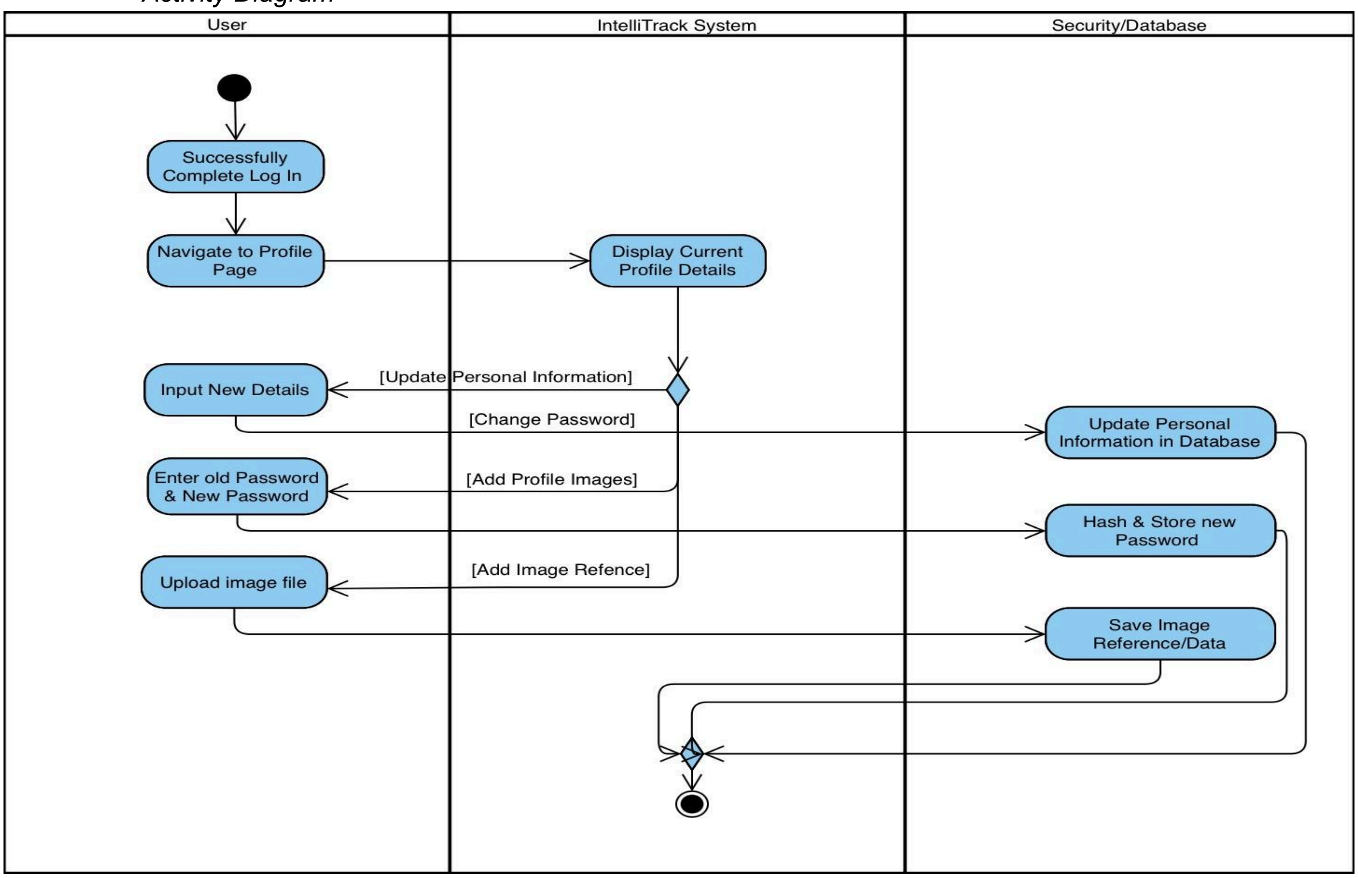


- _Use Case Description_

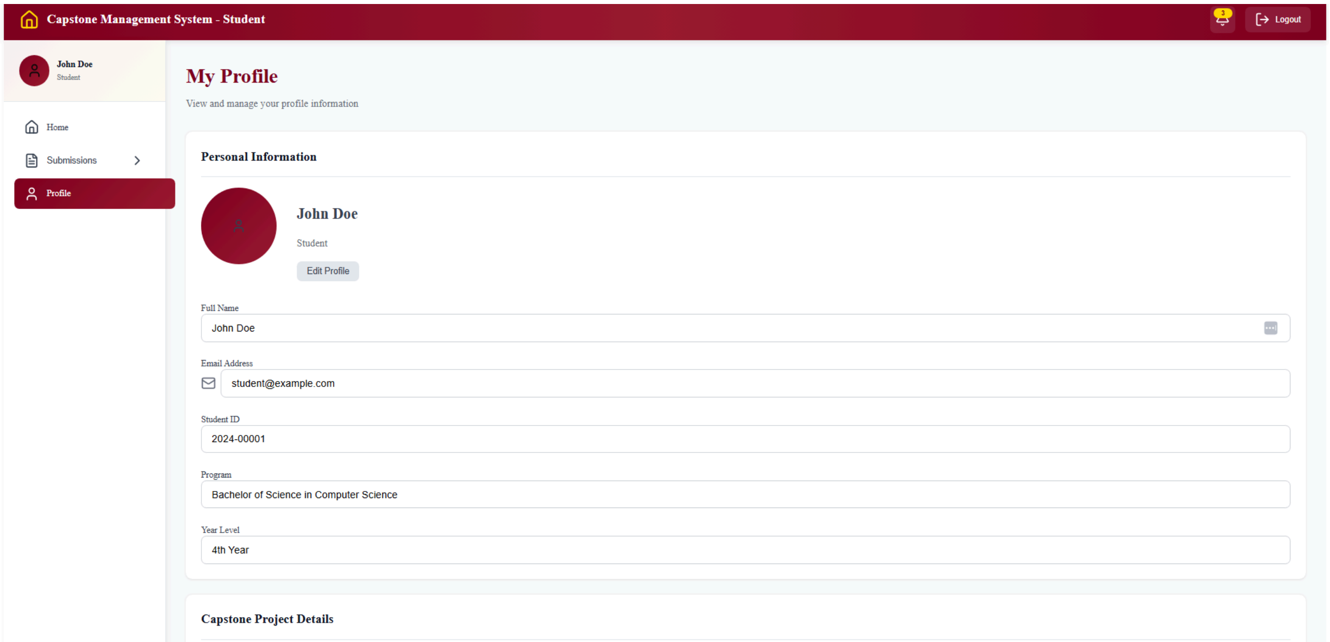
The system shall provide users with the ability to view and manage their personal profile information. Users shall be able to update their personal details, change their password securely, and upload or modify their profile picture.

All updates shall be validated by the system before being stored in the database. Once changes are successfully saved, the updated profile information shall be reflected across the system interface.

- _Activity Diagram_

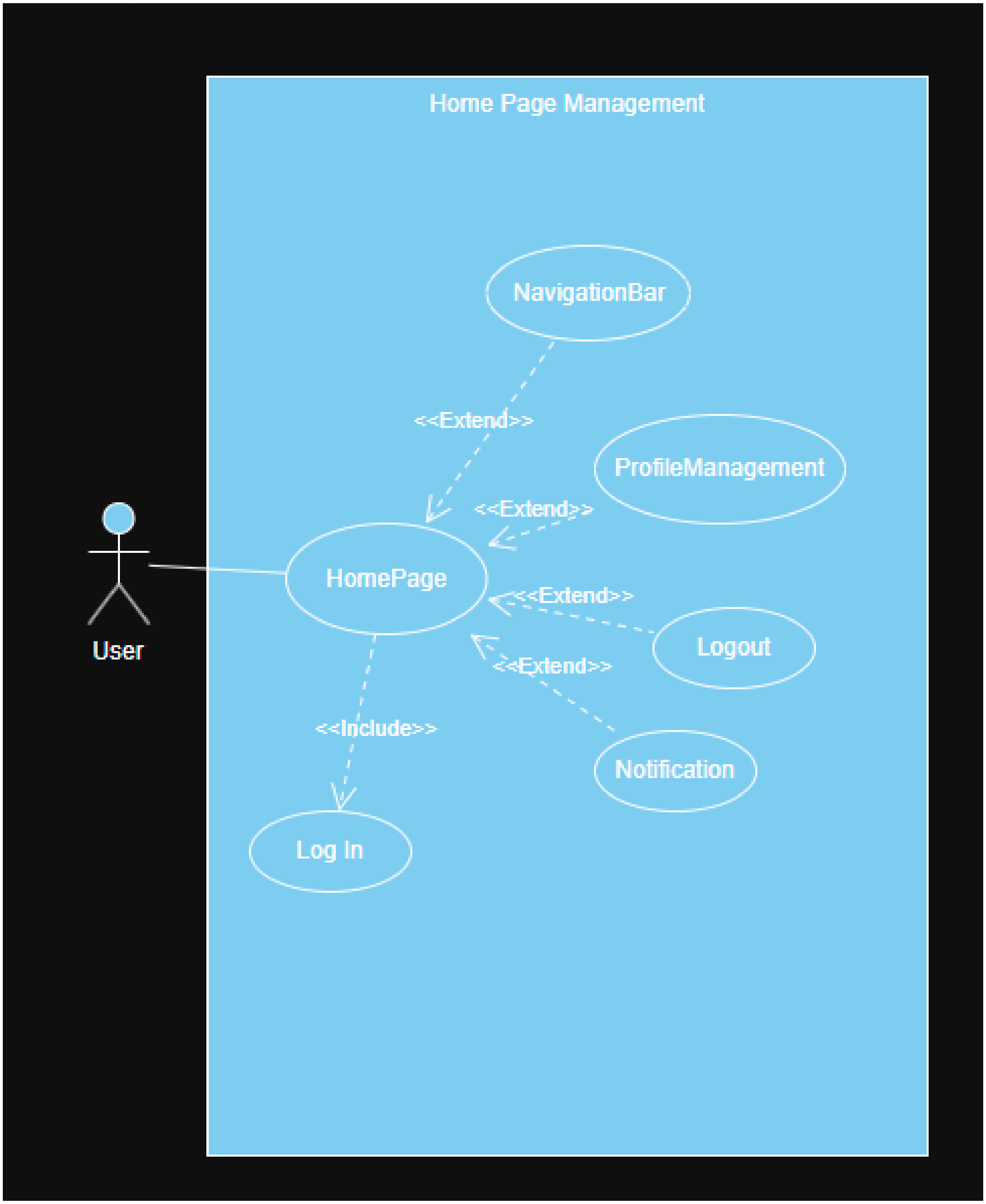


- *Wireframe*


Capstone Management System - Student
Logout
John Doe
Student
Home
Submissions
Profile
My Profile
View and manage your profile information
Personal Information
John Doe
Student
Edit Profile
Full Name
John Doe
Email Address
student@example.com
Student ID
2024-00001
Program
Bachelor of Science in Computer Science
Year Level
4th Year
Capstone Project Details

1.4 **Home Dashboard**

- *Use Case Diagram*

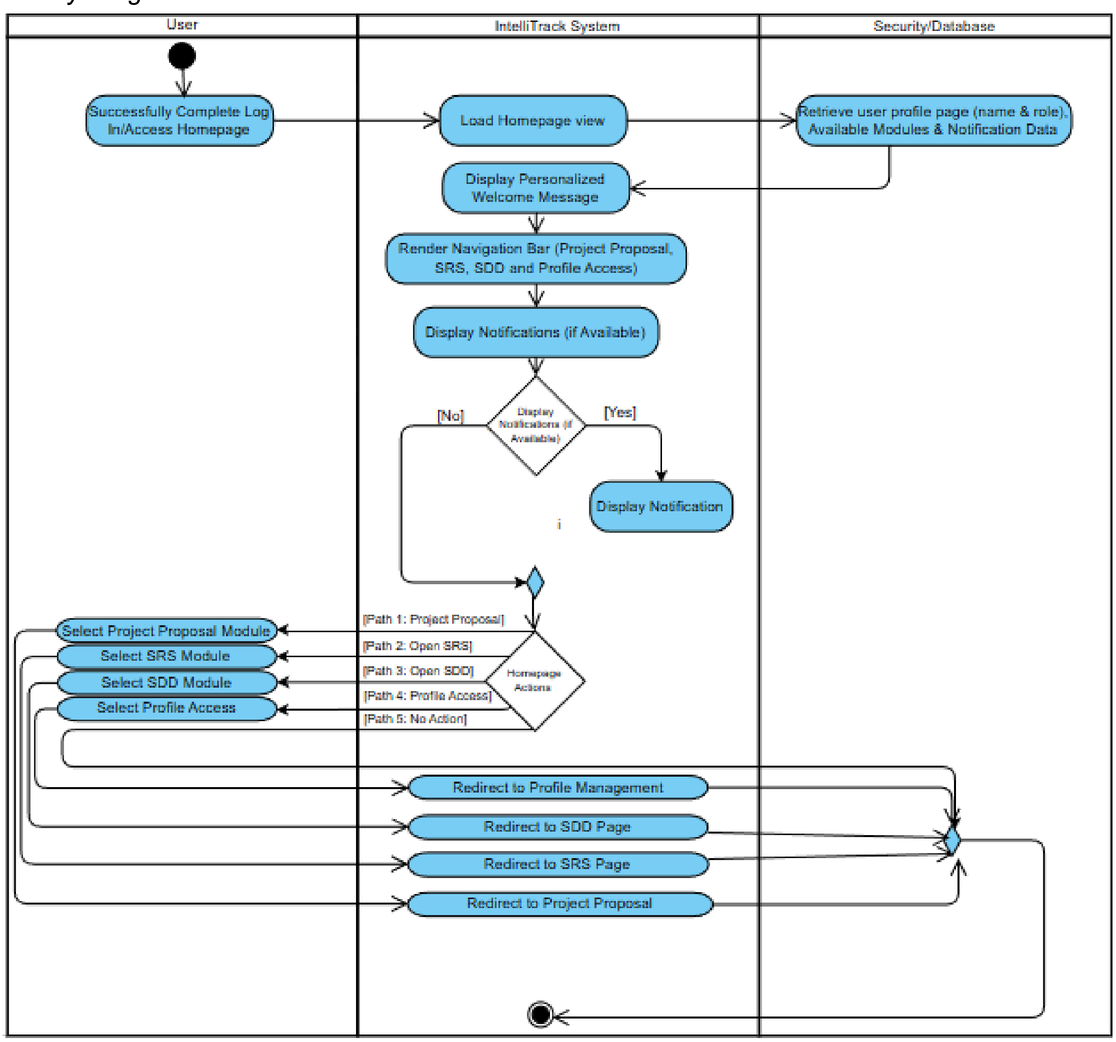


- *Use Case Description*

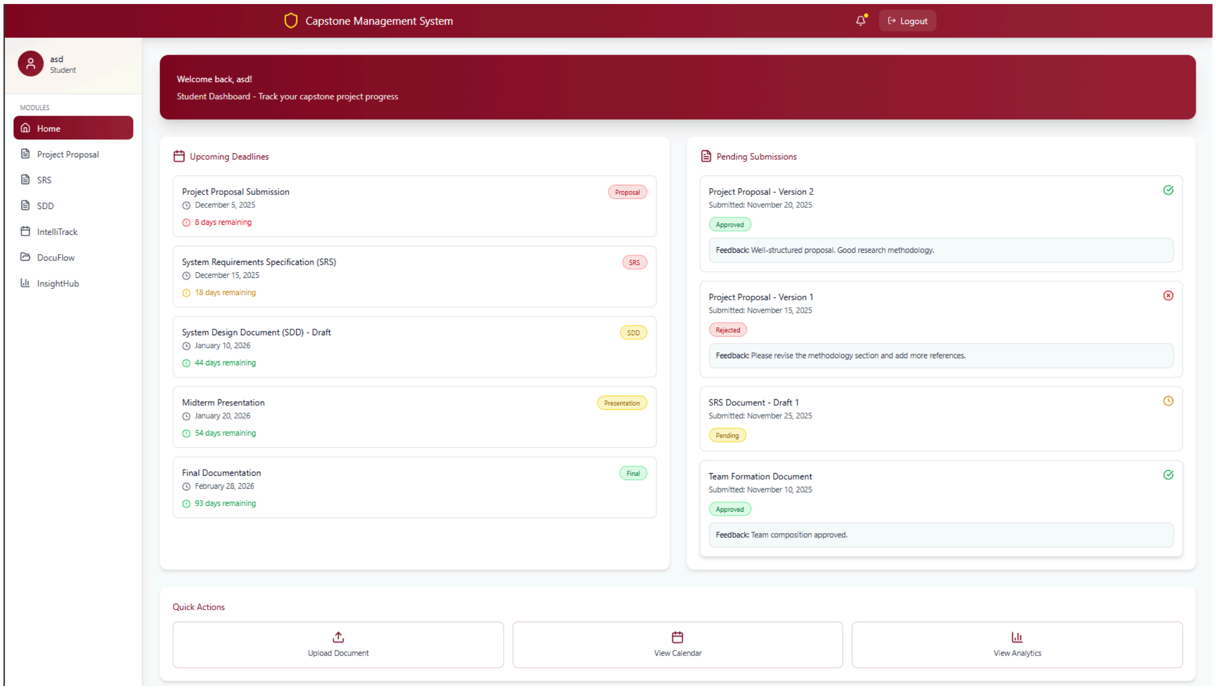
The system shall provide a personalized home page that serves as the main dashboard for users after login. The dashboard shall display the user's name, assigned role, and relevant system information.

The home page shall provide navigation to key modules such as submission tracking, analytics dashboards, and profile management. It shall also display real-time notifications related to deadlines, submission status, and system updates to support user awareness and task management.

- *Activity Diagram*



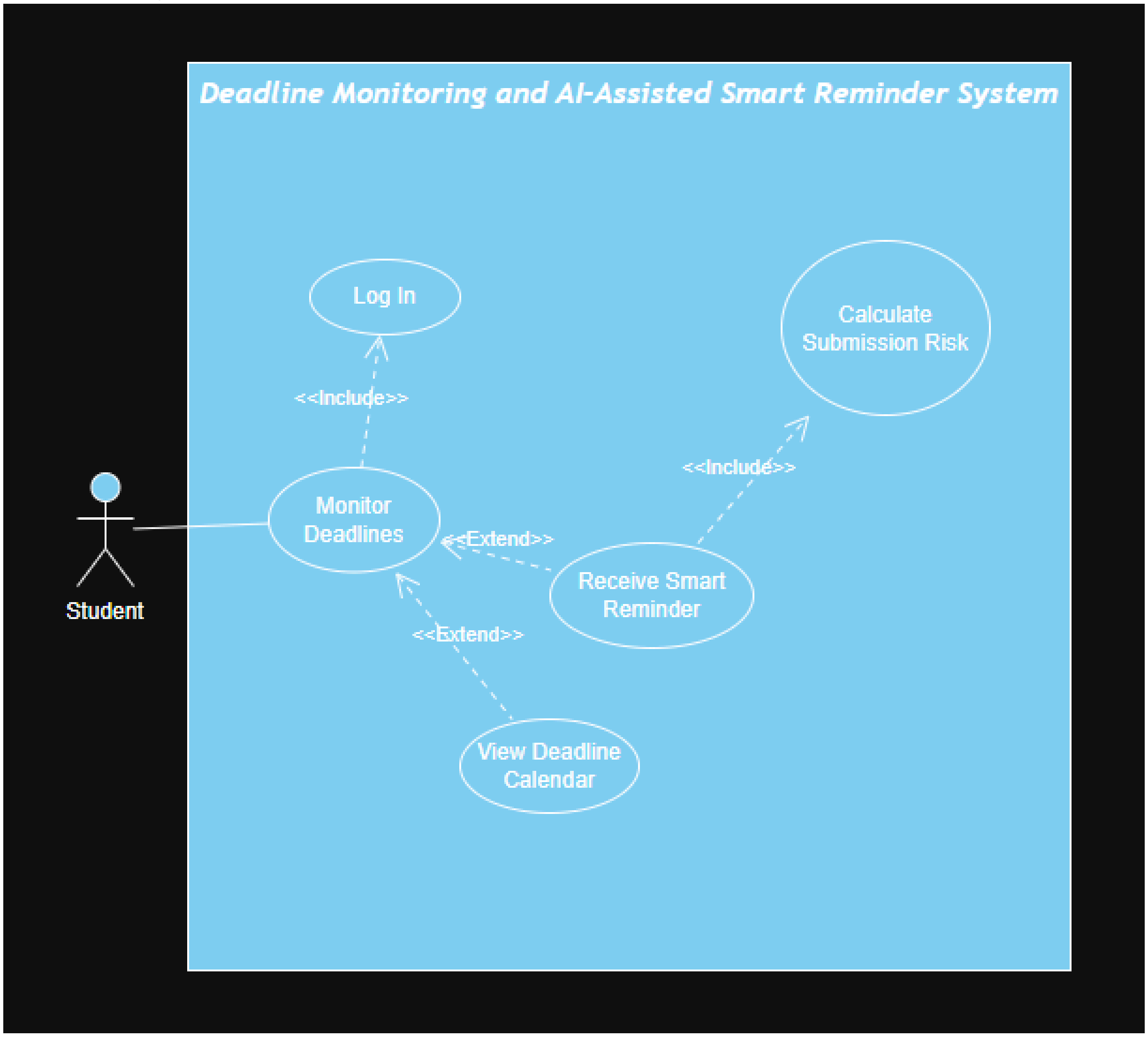
- *Wireframe*


Capstone Management System
Logout
asd
Student
MODULES
Home
Project Proposal
SRS
SDD
IntelliTrack
DocuFlow
InsightHub
Welcome back, asd!
Student Dashboard - Track your capstone project progress
Upcoming Deadlines
Project Proposal Submission
Proposal
December 5, 2025
8 days remaining
System Requirements Specification (SRS)
SRS
December 15, 2025
18 days remaining
System Design Document (SDD) - Draft
SDD
January 10, 2026
44 days remaining
Midterm Presentation
Presentation
January 20, 2026
54 days remaining
Final Documentation
Final
February 28, 2026
93 days remaining
Pending Submissions
Project Proposal - Version 2
Submitted: November 20, 2025
Approved
Feedback: Well-structured proposal. Good research methodology.
Project Proposal - Version 1
Submitted: November 15, 2025
Rejected
Feedback: Please revise the methodology section and add more references.
SRS Document - Draft 1
Submitted: November 25, 2025
Pending
Team Formation Document
Submitted: November 10, 2025
Approved
Feedback: Team composition approved.
Quick Actions
Upload Document
View Calendar
View Analytics

Module 2: Submission Tracker

2.1 ***Deadline Monitoring***

- *Use Case Diagram*

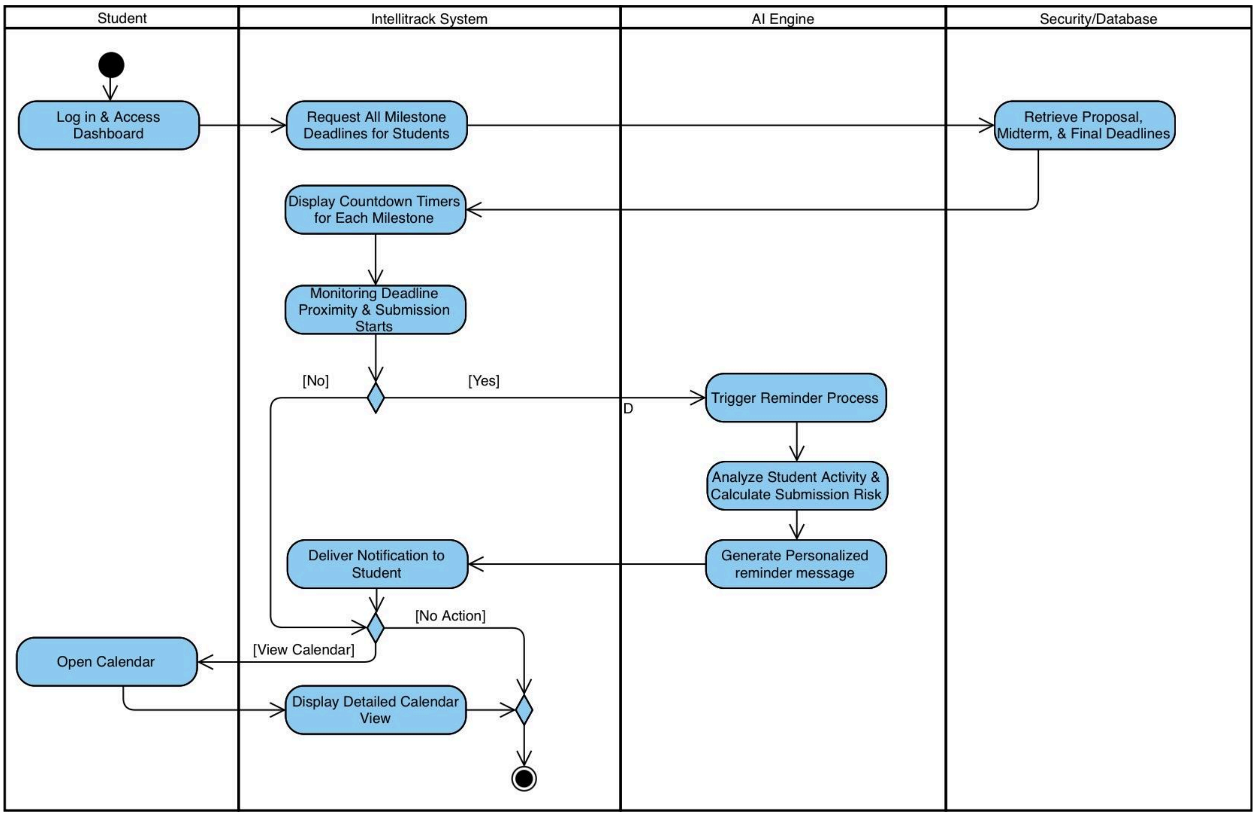


- *Use Case Description*

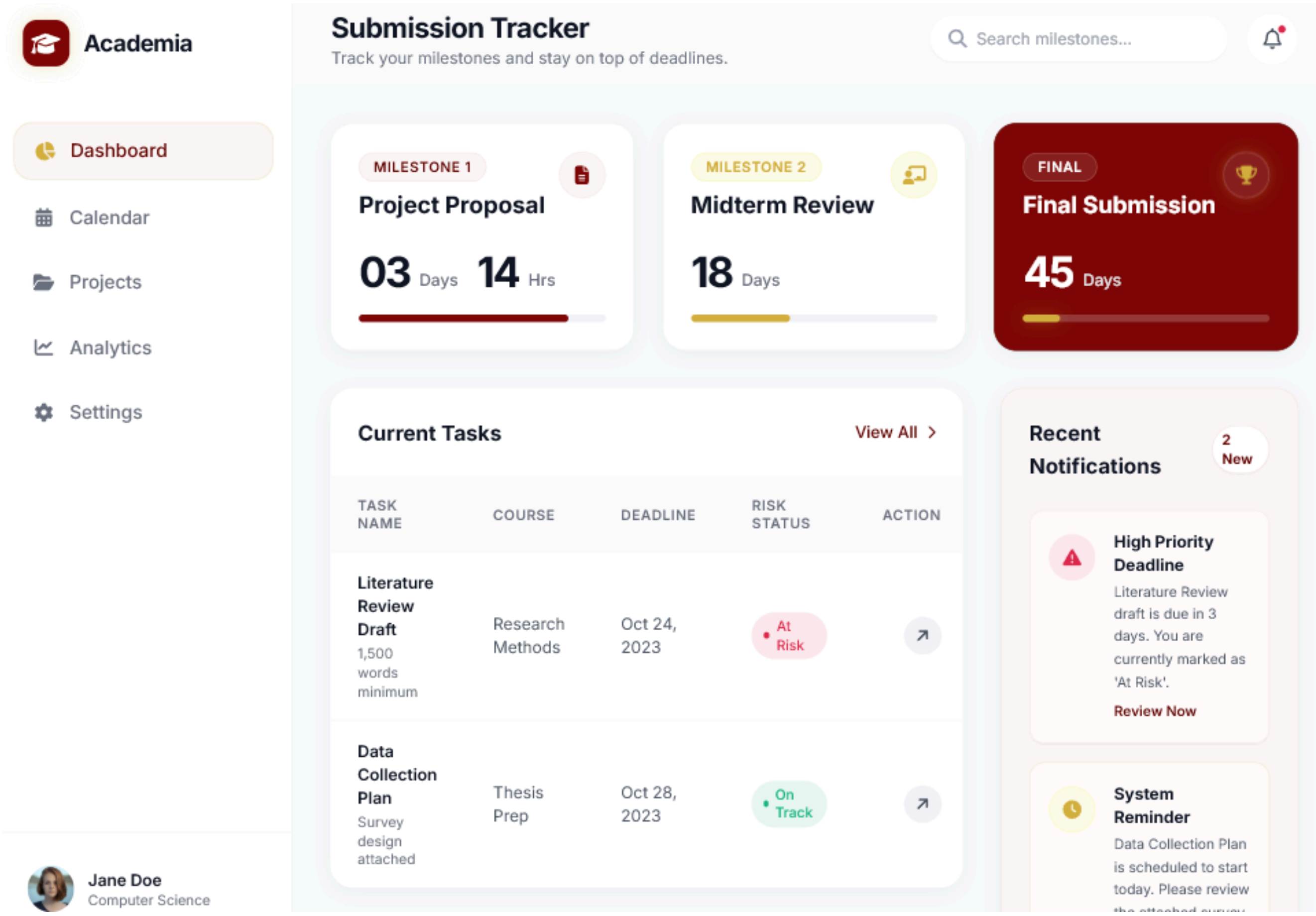
The system shall continuously monitor all configured deliverable deadlines and compare them with the current system time. It shall display countdown timers indicating the remaining time before each submission deadline.

When deadlines are approaching or missed, the system shall generate automated alerts and notifications. AI components shall be used to personalize reminder messages based on user activity and submission behavior to encourage timely completion of deliverables.

- *Activity Diagram*

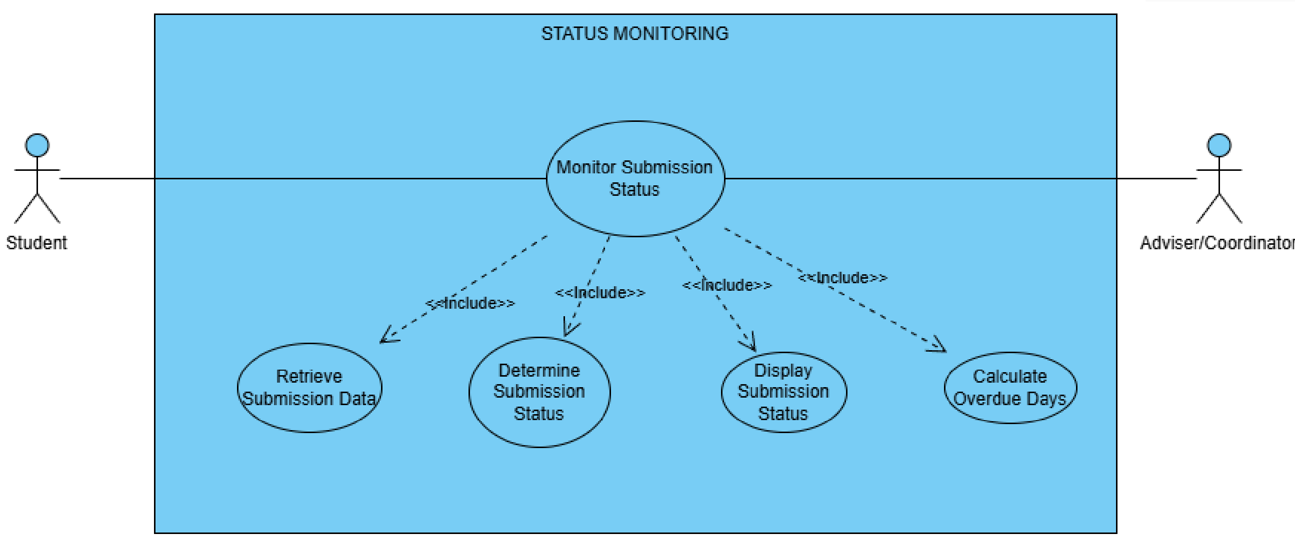


- *Wireframe*



2.2 Status Monitoring

- *Use Case Diagram*

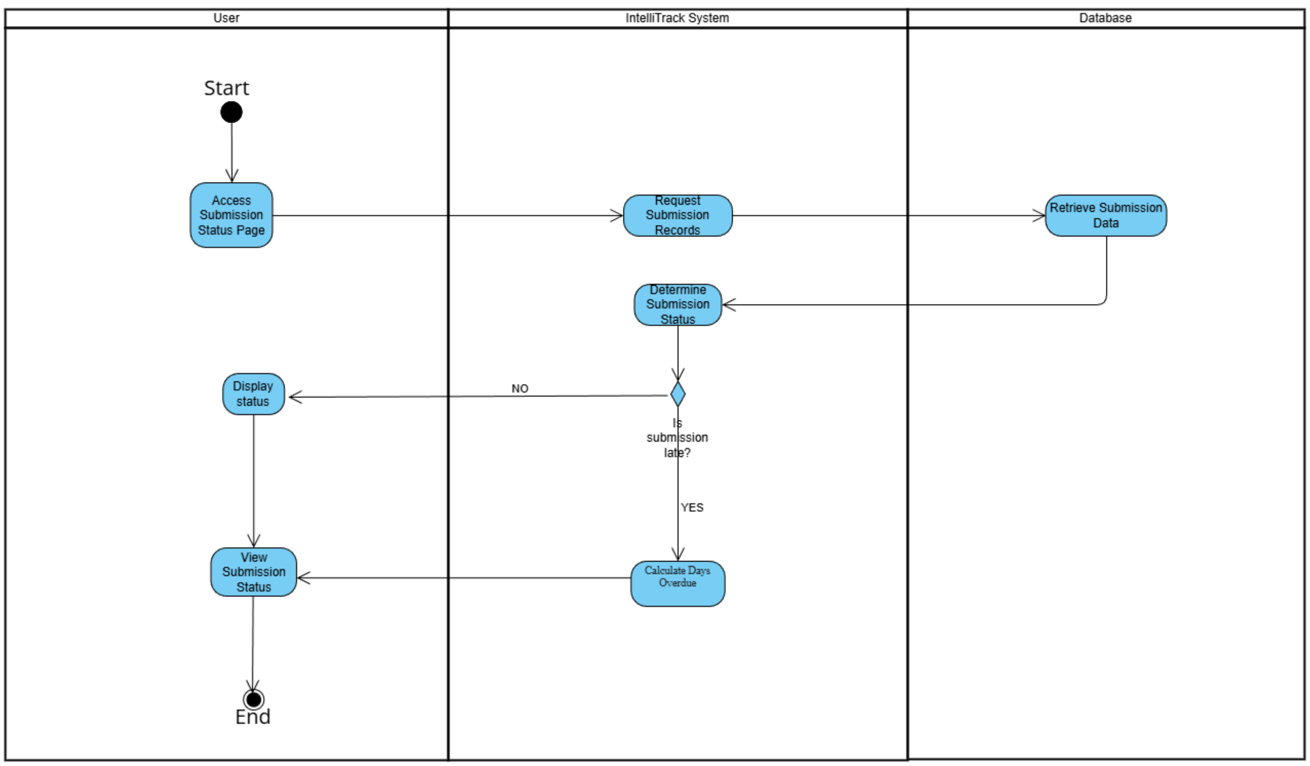


- *Use Case Description*

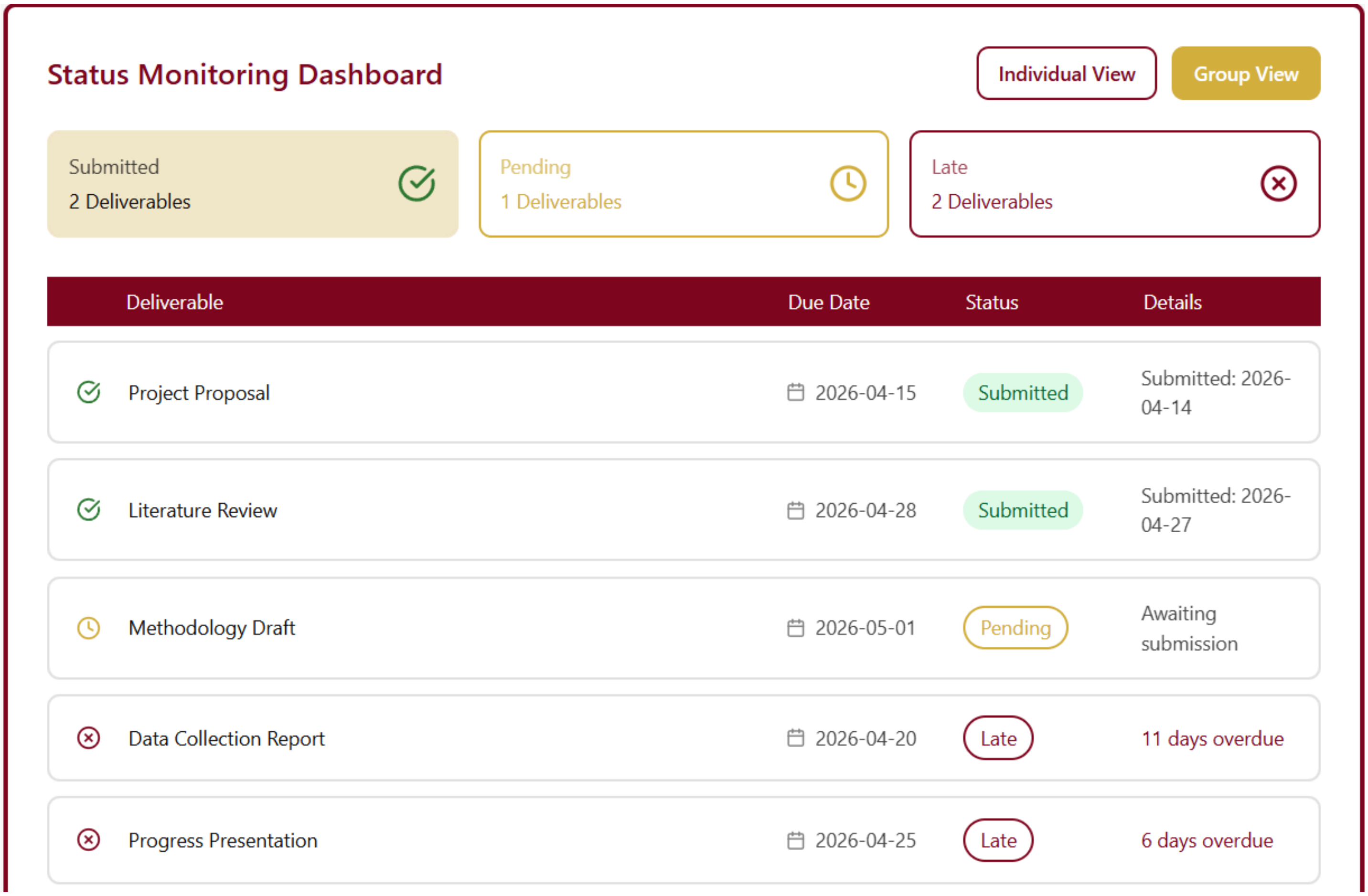
The system shall provide real-time tracking of submission status for each deliverable. It shall retrieve submission records and compare submission timestamps with defined deadlines to determine the current status.

The system shall display whether each deliverable is marked as Submitted, Pending, or Late. For late submissions, the system shall calculate and display the number of overdue days. This feature shall allow users and advisers to monitor submission compliance at both individual and group levels.

- *Activity Diagram*



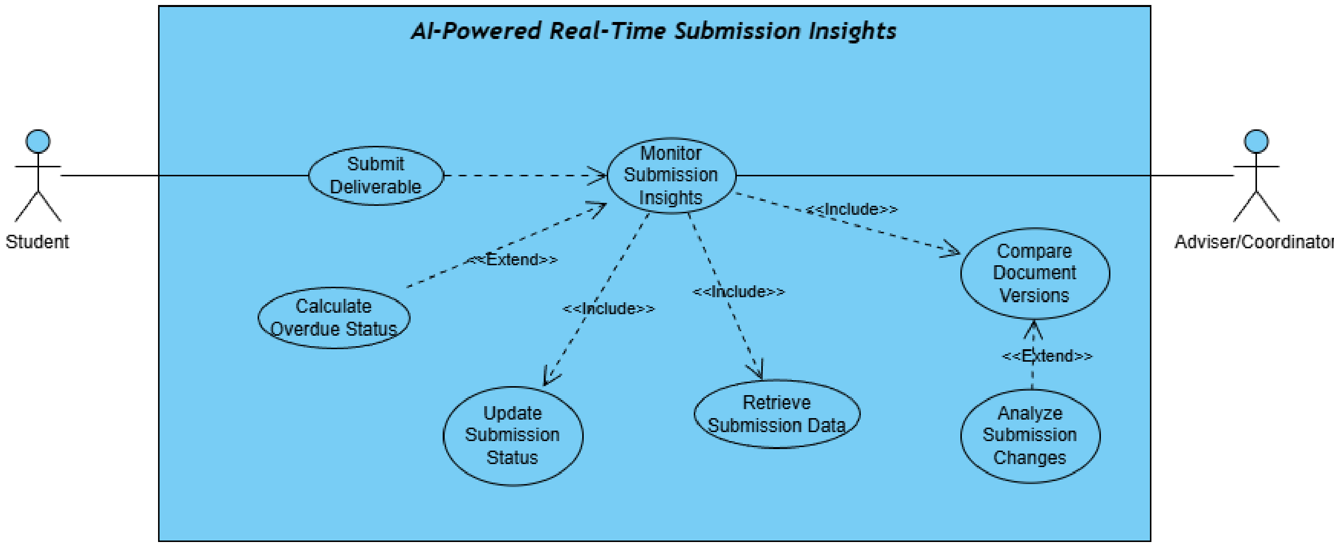
- _Wireframe_


Status Monitoring Dashboard
Individual View
Group View
Submitted
2 Deliverables
Pending
1 Deliverables
Late
2 Deliverables
Deliverable
Due Date
Status
Details
Project Proposal
2026-04-15
Submitted
Submitted: 2026-04-14
Literature Review
2026-04-28
Submitted
Submitted: 2026-04-27
Methodology Draft
2026-05-01
Pending
Awaiting submission
Data Collection Report
2026-04-20
Late
11 days overdue
Progress Presentation
2026-04-25
Late
6 days overdue

Module 3: Analytics & Reporting Dashboard

3.1 _**AI-Powered Real-Time Submission Insights**_

- _Use Case Diagram_



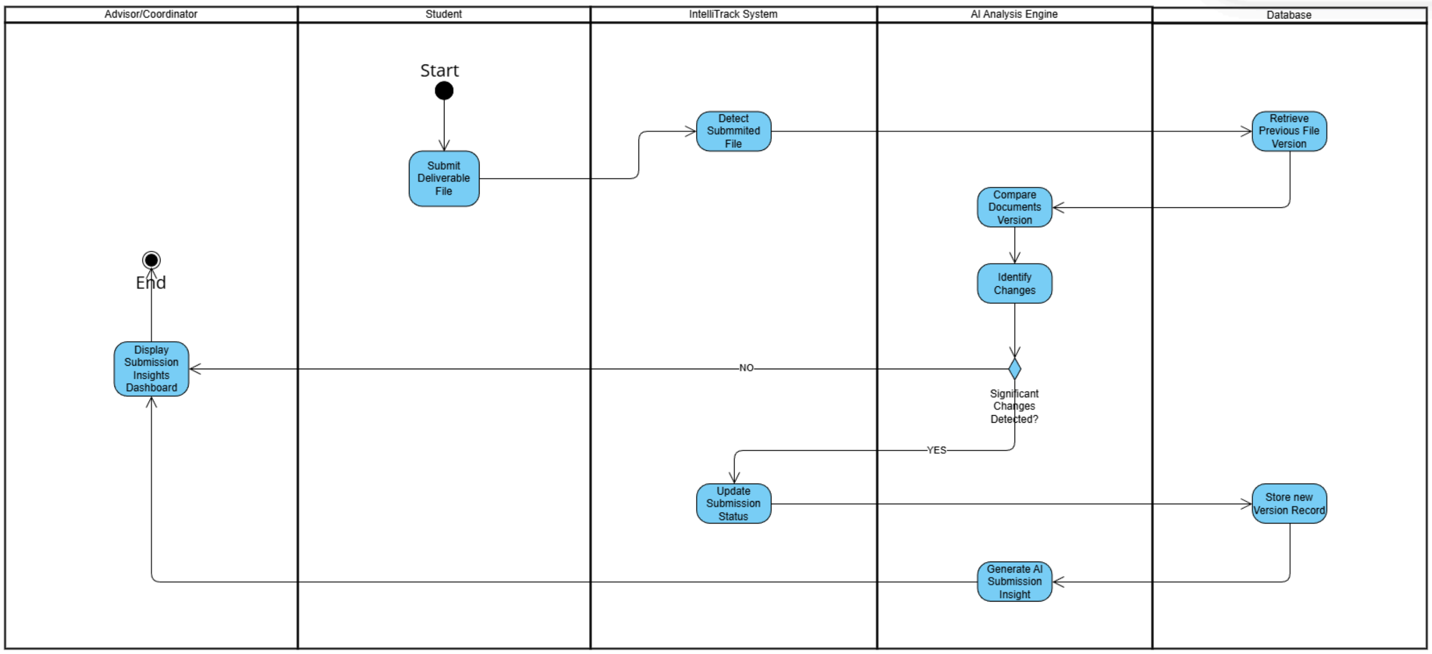
- _Use Case Description_

The system shall provide advisers with an AI-powered dashboard that delivers real-time insights into submission activities. The system shall automatically detect submission events and track changes between document versions.

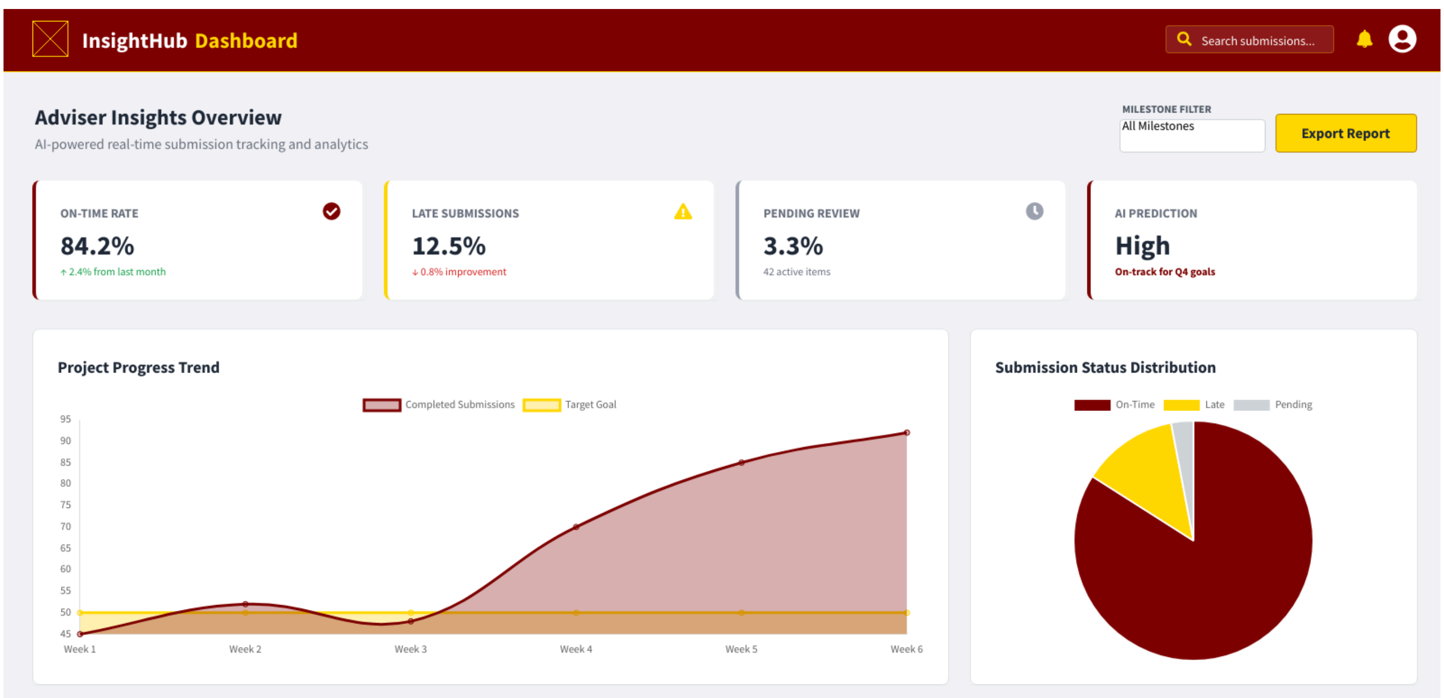
Using AI-assisted comparison, the system shall analyze differences between the latest and previous submissions to identify meaningful updates. Based on this analysis, the system shall update the submission status accordingly and maintain a version history of submitted documents.

The generated insights shall help advisers monitor submission progress and identify significant changes in deliverable updates.

- *Activity Diagram*

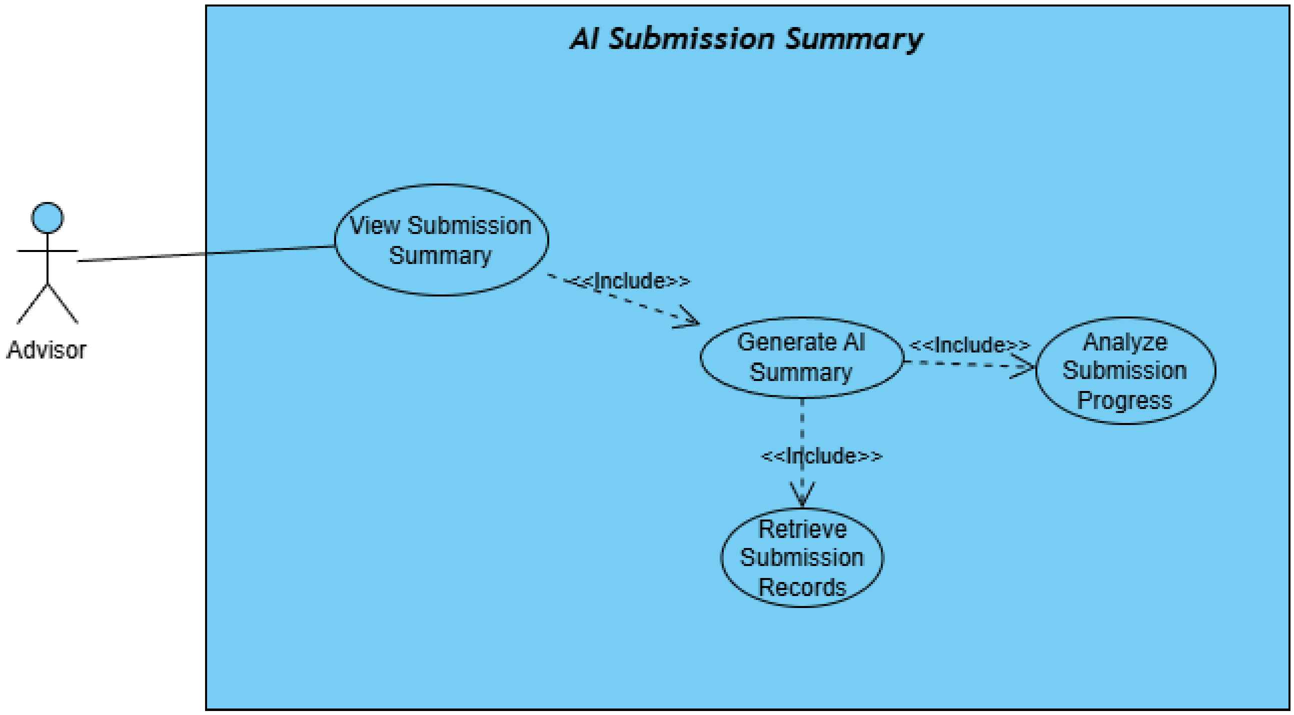


- *Wireframe*



3.2 **AI Submission Summary**

- _Use Case Diagram_

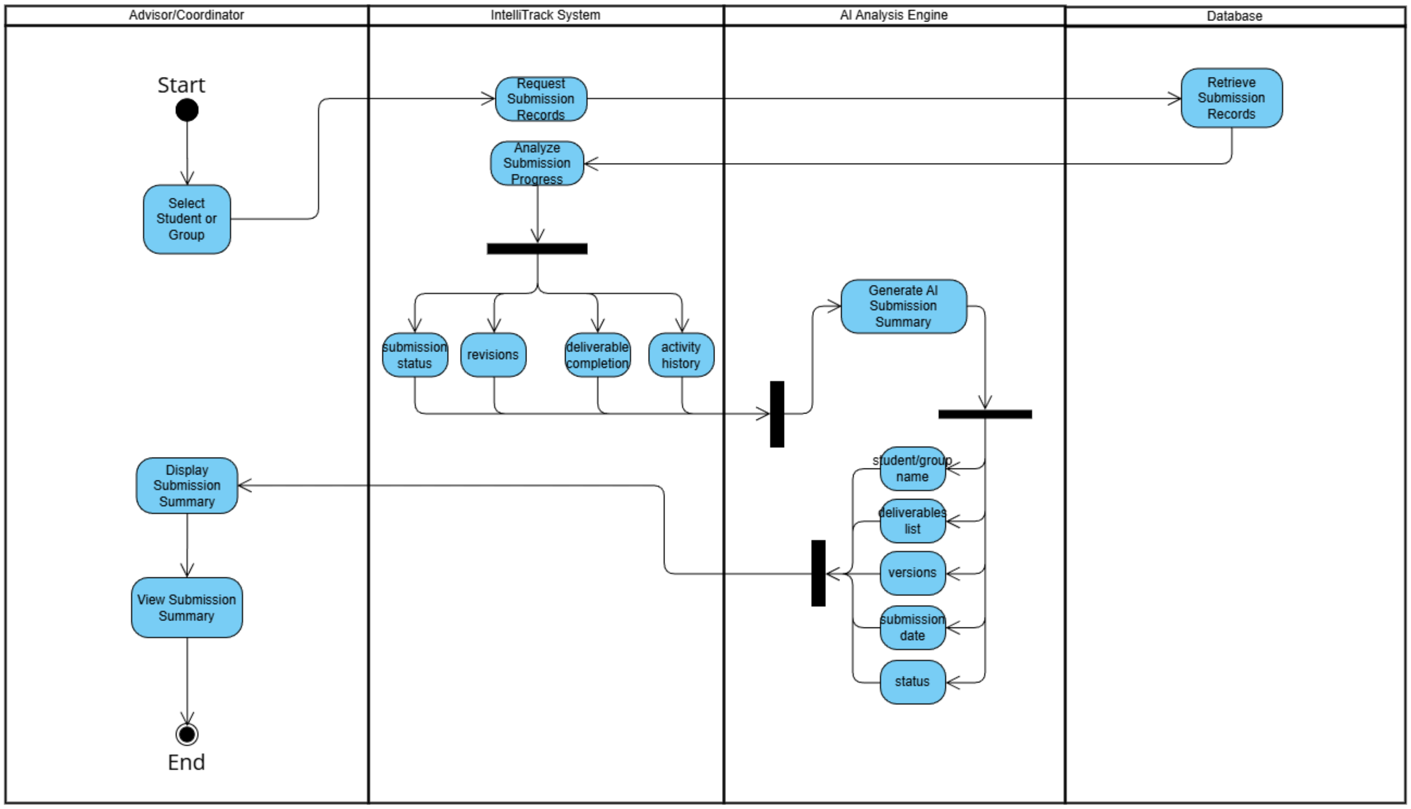


- _Use Case Description_

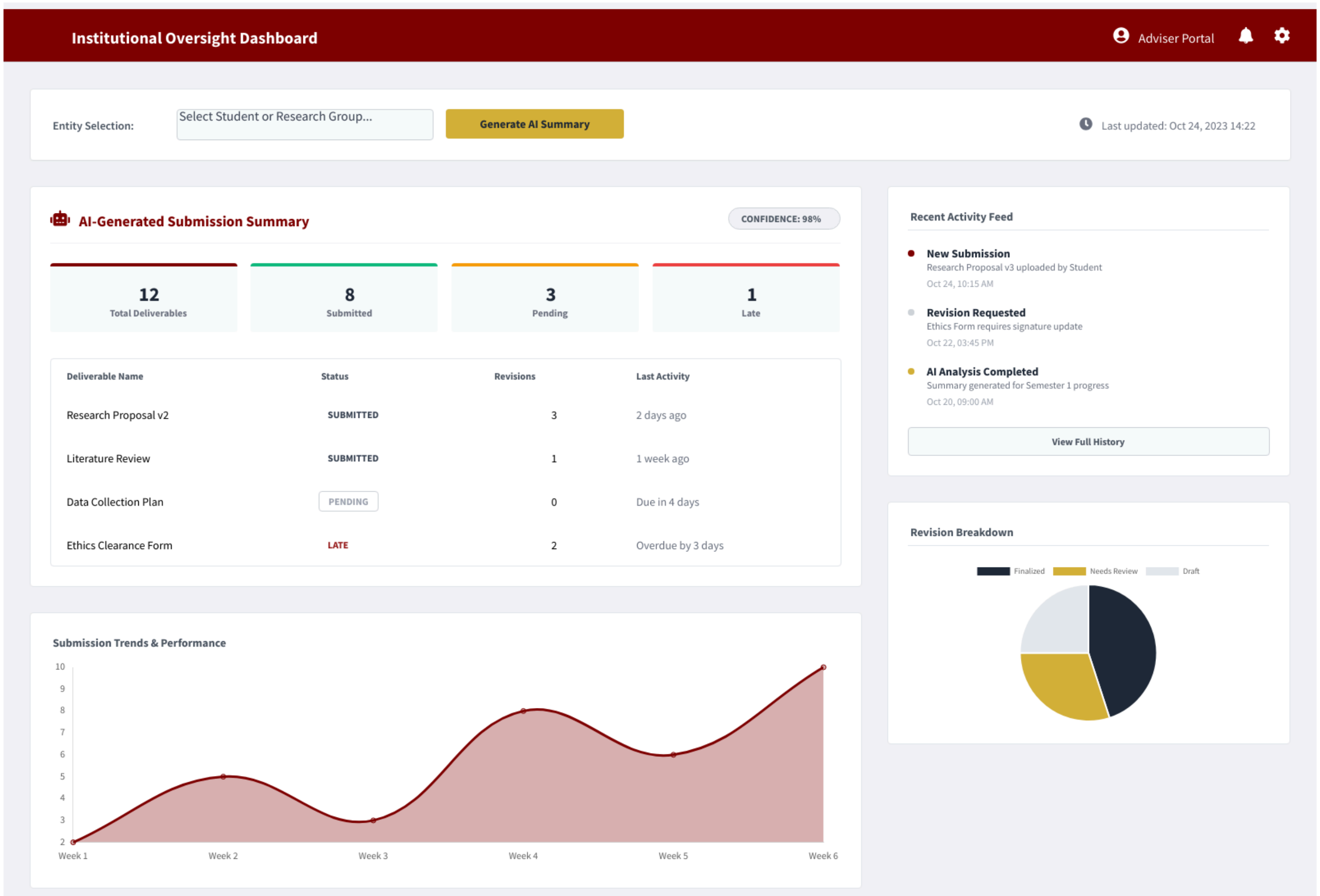
The system shall allow advisers to generate an AI-based summary of submission progress for a selected student or group. Upon selecting a specific entity, the system shall retrieve relevant submission records and analyze the overall progress.

The system shall generate a structured summary that includes key information such as the list of deliverables, submission status, timestamps, revision count, and most recent submission activity. This summary shall provide a concise overview of submission performance, enabling advisers to quickly understand progress without reviewing individual files.

- *Activity Diagram*

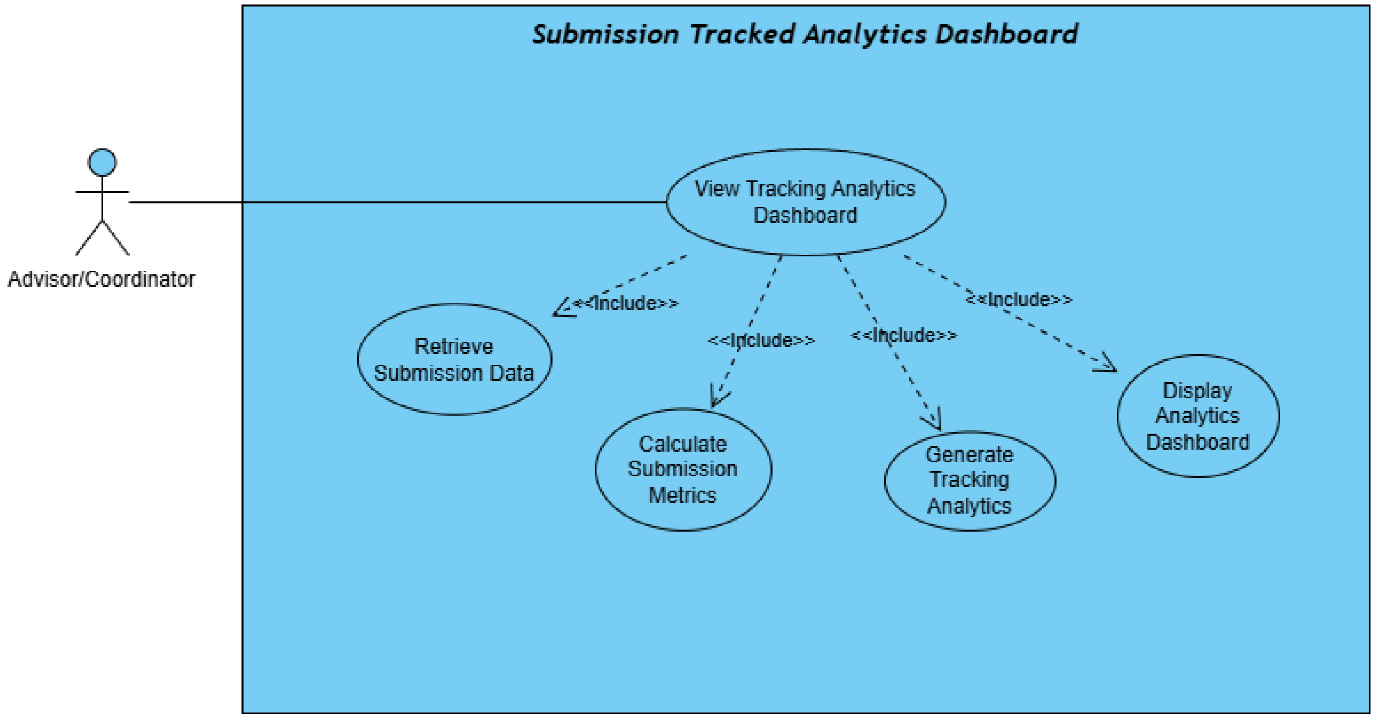


- *Wireframe*


Institutional Oversight Dashboard
Adviser Portal
Entity Selection:
Select Student or Research Group...
Generate AI Summary
Last updated: Oct 24, 2023 14:22
AI-Generated Submission Summary
CONFIDENCE: 98%
12
Total Deliverables
8
Submitted
3
Pending
1
Late
Deliverable Name
Status
Revisions
Last Activity
Research Proposal v2
SUBMITTED
3
2 days ago
Literature Review
SUBMITTED
1
1 week ago
Data Collection Plan
PENDING
0
Due in 4 days
Ethics Clearance Form
LATE
2
Overdue by 3 days
Recent Activity Feed
New Submission
Research Proposal v3 uploaded by Student
Oct 24, 10:15 AM
Revision Requested
Ethics Form requires signature update
Oct 22, 03:45 PM
AI Analysis Completed
Summary generated for Semester 1 progress
Oct 20, 09:00 AM
View Full History
Revision Breakdown
Finalized
Needs Review
Draft
Submission Trends & Performance
10
9
8
7
6
5
4
3
2
Week 1
Week 2
Week 3
Week 4
Week 5
Week 6

*3.3 **Submission Tracking Analytics Dashboard***

- *Use Case Diagram*



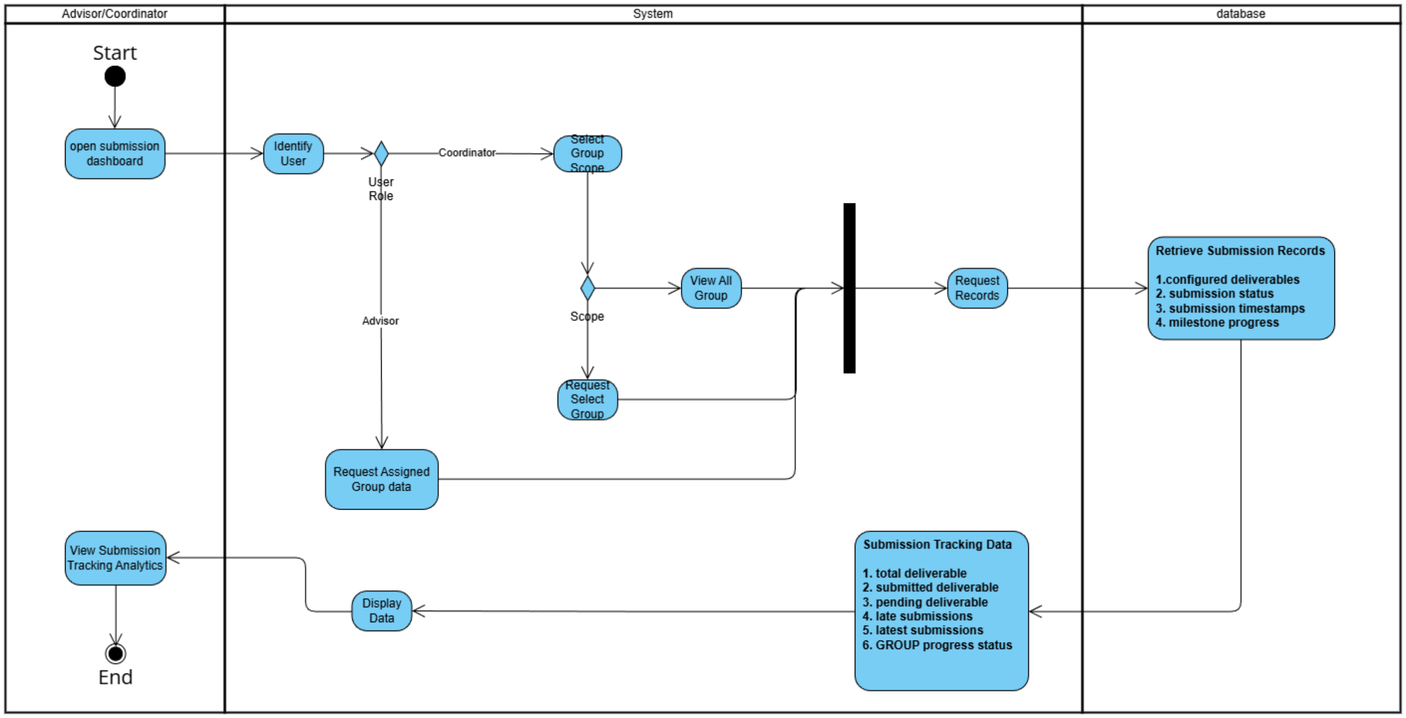
- *Use Case Description*

The system shall provide a tracking analytics dashboard that displays an overview of submission activity and milestone completion. When a user accesses the dashboard, the system shall retrieve submission data and present it in a structured and visual format.

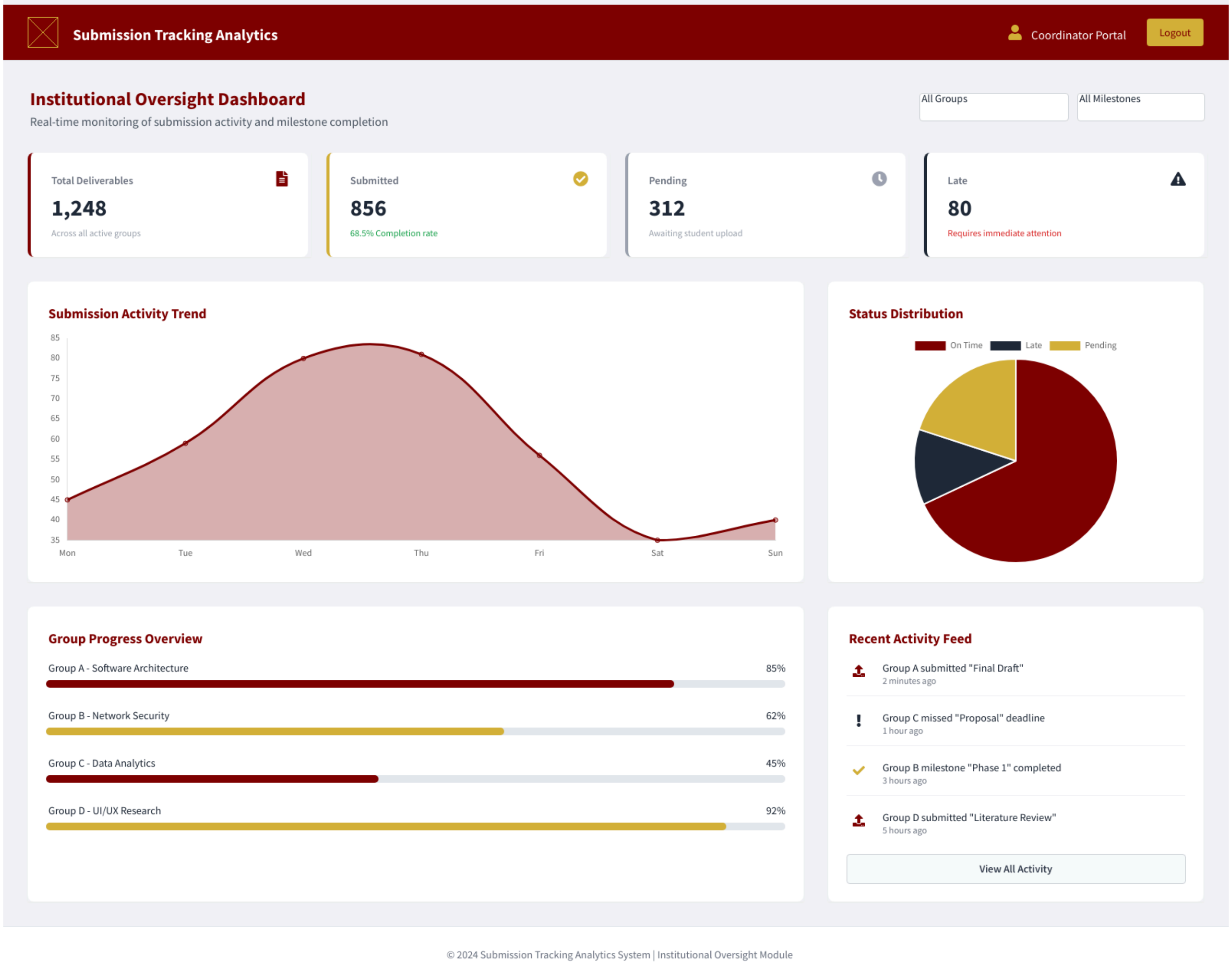
The dashboard shall display key metrics such as total deliverables, submitted deliverables, pending deliverables, late submissions, and recent submission activity. It shall also present the progress status of students or groups based on submission completion.

The system shall ensure that advisers can view analytics for their assigned groups, while coordinators can view analytics for selected groups or all groups. The dashboard focuses on monitoring submission tracking rather than evaluating academic performance.

- *Activity Diagram*



- *Wireframe*



●.4 Non-functional requirements

Performance

- **Response Time:** All major user interactions (login, document upload, dashboard view) must be completed in **less than 2 seconds**.
- **Load Handling:** The system must handle **1,000 concurrent users** during peak usage periods (e.g., near deadline dates) without performance degradation.
- **Scalability:** The cloud-based architecture must support an increase in the number of users, projects, and future modules without requiring a system redesign or causing

performance loss.

- **Real-Time Insights:** Analytics data displayed in the InsightHub must be refreshed in **near real-time** (data latency of less than 5 seconds).
- **Cross-Platform Performance:** System performance must be consistent and reliable across all supported major modern browsers (**Chrome, Edge, Safari, Firefox**) and desktop/mobile devices.

Security

- **Data Encryption:** All documents and sensitive user data must be protected using **end-to-end encryption** (both in transit and at rest).
- **Authentication: Multi-Factor Authentication (MFA)** must be enforced for all **Adviser** and **Administrator** accounts to enhance login security.
- **Access Control:** The system must strictly enforce **Role-Based Access Control (RBAC)** to ensure users can only access data and functions authorized for their specific role.
- **Compliance:** The system must adhere to institutional academic policies and comply with the **Data Privacy Act of 2012 (Philippines)** regarding user data and document handling.
- **Anomaly Detection:** The system must include mechanisms to detect and alert administrators to **suspicious login attempts** or unauthorized file activities.

Reliability

- **System Uptime:** The CDSS must maintain an **availability rate of 99.9%** (Platform Stability), excluding scheduled maintenance.
- **Data Integrity & Recovery:** The system must implement **automated backup and recovery plans** to ensure the integrity, consistency, and restorability of all submissions, feedback, and user data.
- **Auditability: Immutable audit trails** must be maintained for all critical actions (submissions, edits, role changes, and feedback) to ensure academic integrity.
- **Maintainability & UX:** The code base must be **modular, well-documented**, and

follow established standards to ensure stability and ease of bug fixes. The interface must be **simple, intuitive, and clean** (Usability/UX) to minimize operational errors and increase user trust.